Data Mining Project – Milestone 2: Association and Clustering (May 2026)

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***Abstract*—This project presents a comprehensive data mining analysis of a global billionaires dataset comprising 2,600 records and 35 variables, sourced from reliable references such as Forbes, Bloomberg, and official economic databases. Through systematic Exploratory Data Analysis (EDA), the study reveals key patterns in wealth distribution, including the dominance of the United States and China, the prevalence of self-made billionaires, and the concentration of wealth in sectors such as Finance & Investments, Manufacturing, and Technology.**

Extensive data preprocessing was performed, including duplicate removal, hierarchical imputation of missing values, elimination of high-missingness features (e.g., organization, state), standardization of industries and sources into macro-categories, feature engineering (e.g., number of companies), discretization, binarization, and country-level aggregation. These steps prepared the data for advanced mining tasks.

The association task employed transactional representation and rule mining to uncover relationships between individual attributes (age, industry, source of wealth, self-made status) and country-level socioeconomic indicators (GDP, education, life expectancy). The clustering and classification components focused on identifying distinct national wealth ecosystems: Innovation Ecosystem, Rentier Paradise, and Extraction Economy, based on sectoral composition, self-made ratios, demographic structure, and economic indicators. The results provide actionable insights into the structural drivers of global extreme wealth and their relation to national economic models.

***Index Terms*—Data Mining, Exploratory Data Analysis (EDA), Data Preprocessing, Data Cleaning, Feature Engineering, Dimensionality Reduction, Association, Clustering.**

I. Introduction

Extreme wealth concentration is one of the most salient phenomena of contemporary global economics. Billionaires, as the apex of wealth distribution, serve not only as indicators of individual success but also as proxies for the underlying economic structures, innovation capacity, institutional quality, and sectoral dynamics of the countries in which they operate.

This project, part of the Data Mining course, leverages a rich dataset of approximately 2,600 billionaires worldwide to conduct an in-depth analysis of wealth generation patterns. The dataset integrates personal characteristics (age, gender, self-made status, industry, source of wealth) with country-level macroeconomic and demographic indicators (GDP, population, education enrollment, life expectancy, tax rates, etc.).

The main objectives of this milestone are:

Perform thorough Exploratory Data Analysis (EDA) to understand distributions, identify anomalies, and uncover initial insights.

Apply rigorous data preprocessing techniques, including cleaning, imputation, feature engineering, discretization, and transformation, to prepare the data for mining tasks.

Execute Association Rule Mining to discover meaningful relationships between billionaire attributes and national economic conditions.

Develop clustering and classification models at the country level to identify distinct national "wealth ecosystems": Innovation-driven, Rentier (asset/inheritance-based), and Extraction-oriented economies.

By combining individual-level and aggregated country-level perspectives, this study aims to move beyond descriptive statistics and reveal structural patterns that explain how different societies generate and sustain extreme wealth. The ultimate goal is to contribute to a deeper understanding of the interplay between personal entrepreneurship, sectoral specialization, and macroeconomic environments in the formation of global billionaires.

II. Dataset description

A. Dataset

The dataset contains comprehensive information about billionaires around the world, compiled from multiple reliable sources, including official reports, government databases, financial institutions, and well-known publications such as Forbes, Bloomberg, and the World Bank.

The dataset covers several areas of analysis, including:

- Wealth distribution
- Demographic trends
- Economic indicators
- Geographic information and country coordinates

The data was collected using a combination of manual data collection and automated extraction techniques. Researchers and analysts gathered information from official reports, publications, and trusted websites. Additionally, tools such as web scraping and APIs were used to extract structured data from online sources.

This approach allowed the creation of a comprehensive and structured dataset, suitable for exploratory data analysis and data mining studies.

B. Size

The dataset has 2600 rows and 35 columns

C. Dataset Variables Description

There are 35 columns. We are going to describe them on Table 2.1.

TABLE 2.1
Summary of all variables

Variable	Type	Description
rank	int64	Position of the individual in the global billionaire ranking
finalWorth	int64	Net worth of the individual expressed in monetary value
category	object	Classification of the individual (e.g., entrepreneur, investor)

Variable	Type	Description
personName	object	Full name of the individual
age	float64	Age of the individual
country	object	Country associated with the individual
city	object	City of residence or origin
source	object	Main source of wealth
industries	object	Industry or sector associated with the wealth
countryOfCitizenship	object	Country of citizenship
organization	object	Organization or company associated with the individual
selfMade	bool	Indicates whether the wealth is self-made or inherited
status	object	D represents self-made billionaires and "U" represents inherited wealth
gender	object	Gender of the individual
birthDate	object	Date of birth of the individual
lastName	object	Last name or surname of the individual
firstName	object	First name of the individual
title	object	Honorific title used by the person (e.g., Mr., Mrs., Dr.)
date	object	Date associated with the dataset entry
state	object	State or region associated with the individual
residenceStateRegion	object	Region of residence
birthYear	float64	Year of birth
birthMonth	float64	Month of birth
birthDay	float64	Day of birth
cpi_country	float64	Consumer Price Index of the country
cpi_change_country	float64	Change in Consumer Price Index
gdp_country	object	Gross Domestic Product of the country
gross_tertiary_education_enrollment	float64	Gross tertiary education enrollment rate
gross_primary_education_enrollment_country	float64	Gross primary education enrollment rate
life_expectancy_country	float64	Average life expectancy in the country
tax_revenue_country_country	float64	Tax revenue collected by the country
total_tax_rate_country	float64	Total tax rate applied in the country
population_country	float64	Total population of the country
latitude_country	float64	Latitude coordinate of the country
longitude_country	float64	Longitude coordinate of the country

From **Table 2.1**, we observe the distribution of variable types are shown in **Figure 2.1**.

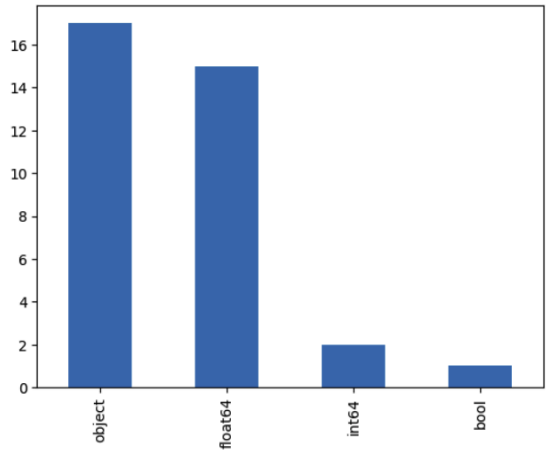


Figure 2.1. Histogram of Dataset Variable Types.

We can observe that there are many descriptive variables and numeric variables. However, there are relatively few integer and boolean variables.

III. Exploratory analysis

Exploratory Data Analysis (EDA) is a fundamental step in the data mining process, aimed at understanding the main characteristics of the dataset before applying any transformations or models. In this phase, the data is examined to identify distributions, detect missing values, analyze variable relationships, and uncover potential inconsistencies or anomalies. This initial analysis provides valuable insights that guide subsequent preprocessing decisions and ensure a more effective and reliable data mining workflow.

A) Null values

We generated a histogram showing the distribution of missing values in the dataset. It can be seen on Figure 4.1 that from this analysis, we observed that four columns contain the highest number of missing values: **organization, title, residenceStateRegion, and state**.

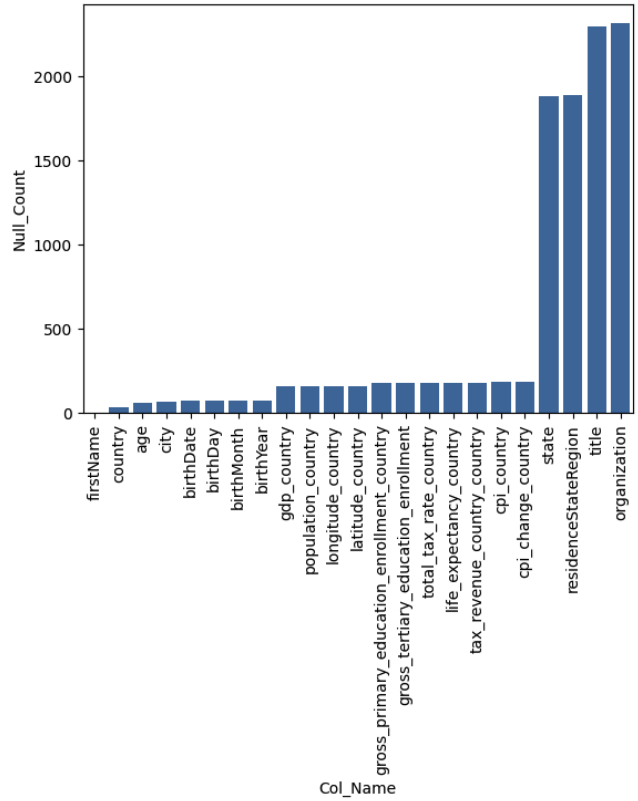


Figure 3.1. Histogram of null values on the dataset.

B) Analysis of categorical variables

We analyzed each of the categorical columns (18 in total). From this analysis, we observed a total of 78 different countries, 741 unique cities, and 906 distinct sources of wealth, spanning more than 18 industries.

From Figure 4.2, during the analysis of the "**category"** variable, we observed that the industries where billionaires are most concentrated are **Finance & Investments, Manufacturing, Technology, Fashion & Retail, and Food & Beverage**. This suggests that these sectors tend to generate the largest fortunes, likely due to their global market reach, high scalability, and strong economic influence.

Additionally, from Figure 4.3, while analyzing the **personName variable**, we observed that the names of individuals are not particularly relevant for the analysis. However, two repeated names were id
entified in the dataset. Therefore, it is necessary to verify whether they correspond to the same individual. If they do, one of the records should be removed to avoid duplicate entries.

Besides, from Figure 4.4, while analyzing the **country variable**, we observed that the **United States** has the largest number of billionaires (more than 700), followed by **China** (more than 500) and **India** (more than 150).

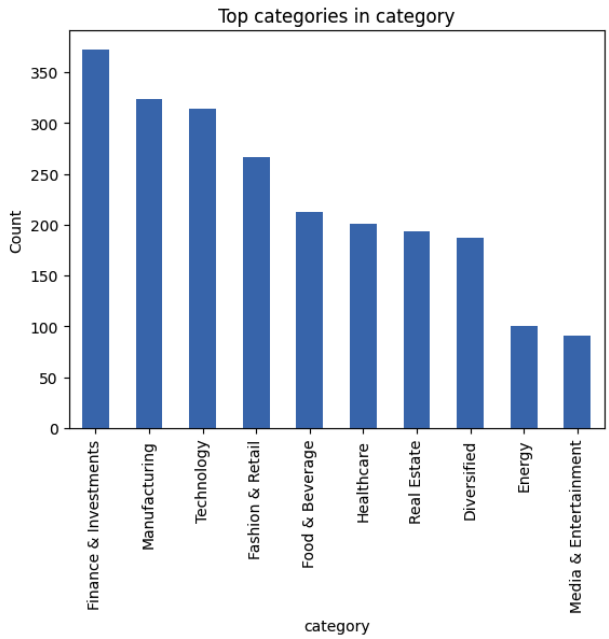


Figure 3.2. Distribution of the top 10 billionaire categories.

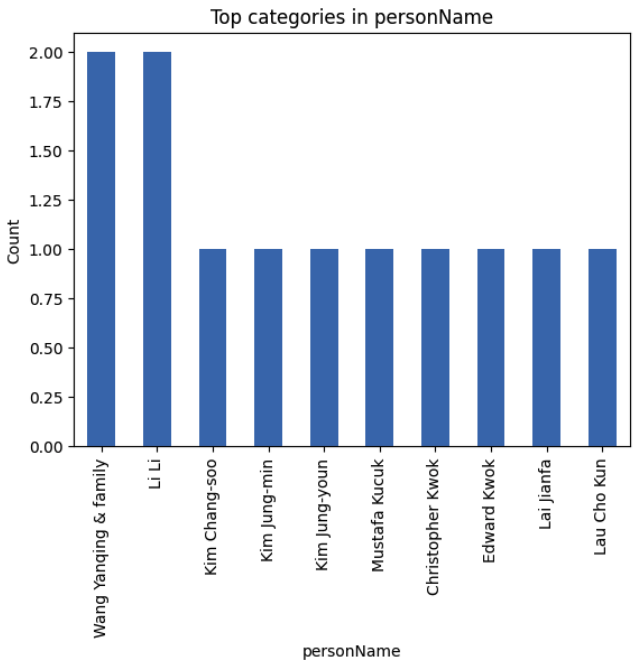


Figure 3.3. Distribution of the top 10 billionaire names.

This pattern may be related to factors such as **population size and the scale of their economies**. The United States and China have the largest economies in the world in terms of GDP, which creates more opportunities for large companies, investments, and wealth generation. Larger economies typically support more industries, financial markets, and technological development, all of which contribute to the creation of billionaires.

In addition, China and India have extremely large populations, which increases the number of potential entrepreneurs, investors, and business leaders. With larger populations and expanding economies, the probability of generating very high levels of wealth also increases.

Therefore, the high number of billionaires in these countries may reflect a combination of **economic size, population scale, and strong business ecosystems**, rather than being solely the result of bias in the dataset.

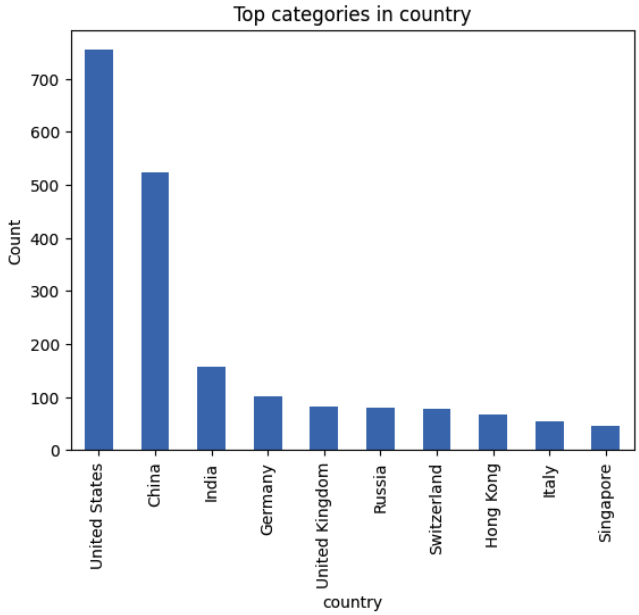


Figure 3.4. Distribution of the top 10 billionaire countries.

Furthermore, from Figure 4.5, while analyzing the city variable using a histogram, we noticed that the cities where billionaires are located tend to be global financial and economic centers. This suggests that billionaires are more likely to live in places where there are investment opportunities, major businesses, and luxury services. Additionally, the concentration of wealth appears to be focused in cities with advanced infrastructure, strong business networks, well-developed legal and financial services, and large economic markets.

Moreover, from Figure 4.6, while analyzing the source variable using a histogram, we observed that real estate leads with more than 140 occurrences. This indicates that many billionaires accumulate their wealth through property, highlighting real estate as a major generator of wealth. Investments and diversified sources follow, suggesting that another large group of billionaires builds their fortunes through financial investments or multiple business activities, spreading risk across different sectors. Additionally, industries such as pharmaceuticals and software, although appearing less frequently, represent high-value sectors that also produce billionaires, reflecting the importance of technological innovation and healthcare in modern wealth creation. Overall, this distribution suggests that while real estate and traditional investments remain dominant, specialized high-value sectors also play an important role in the creation of large fortunes.

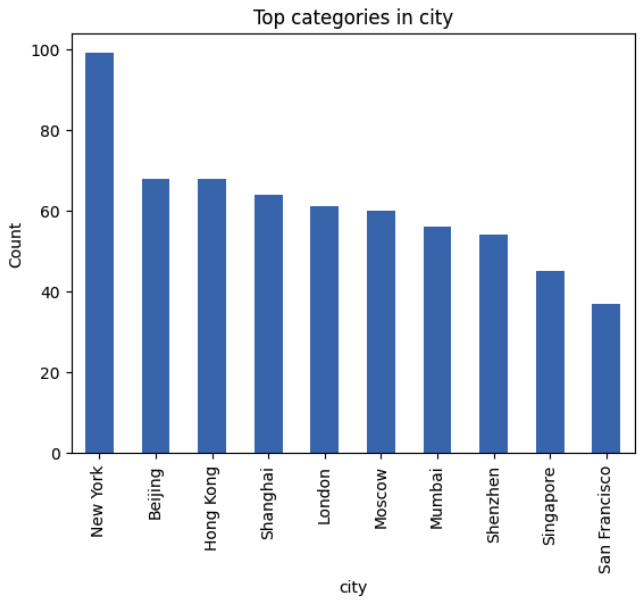


Figure 3.5. Histogram showing the distribution of the top cities with the highest number of billionaires.

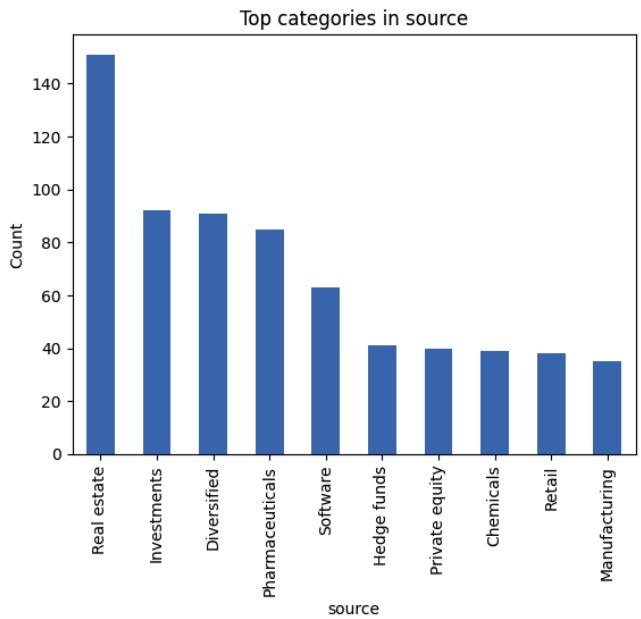


Figure 3.6. Histogram showing the distribution of the most common sources of billionaire wealth.

Similarly, from Figure 4.7, while analyzing the industries variable using a histogram, we observed that Manufacturing has the highest number of occurrences (more than 320), followed by Technology (around 310) and Finance & Investments (more than 250). After these sectors, Fashion & Retail also shows a significant presence with more than 270 occurrences.

This distribution suggests that billionaire wealth is strongly associated with large-scale industries, technological innovation, and financial markets, which tend to generate high economic value and global business opportunities.

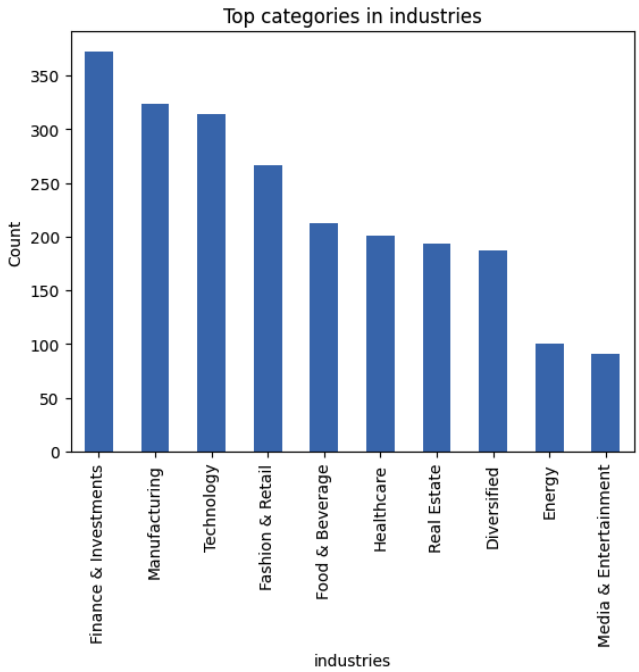


Figure 3.7. Histogram showing the distribution of the most common industries among billionaires

Additionally, it is important to distinguish between the variables source and industries when analyzing billionaire data. The source variable refers to the primary origin of a billionaire's wealth, representing the single most significant contributor to their fortune, such as real estate or software. In contrast, the industries variable lists all sectors in which the individual operates or has business interests, which may include multiple areas such as finance, technology, or manufacturing.

This distinction explains the difference observed in the histograms: while real estate appears as the most common source of wealth, sectors like Finance & Investments, Manufacturing, and Technology appear more frequently in the industries variable because billionaires often have diversified business activities across several sectors.

Therefore, when interpreting these variables, it is important to consider that source represents the main origin of wealth, whereas industries reflects the broader economic involvement of each individual.

Additionally, from Figure 3.8, while analyzing the countryOfCitizenship variable using a histogram, we observed a trend indicating that billionaires generally reside in the country of which they are citizens. While the country variable represents the current country of residence and countryOfCitizenship represents legal citizenship, the histograms show very similar patterns, with the United States, India, and China appearing as the leading countries.

This suggests that most billionaires tend to live in their home country rather than relocating abroad. However, it is important to note that these two variables represent different concepts and may reveal interesting patterns for individuals who reside outside their country of citizenship.

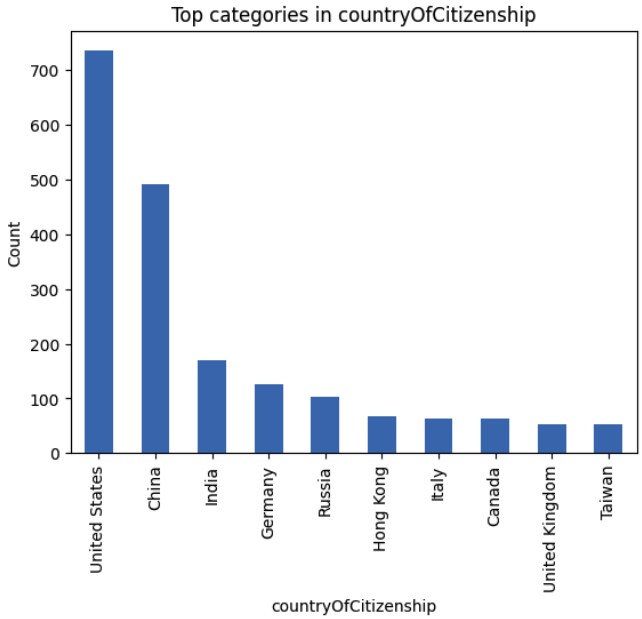


Figure 3.8. Histogram showing the distribution of billionaires by country of citizenship.

Also, from Figure 3.9, while analyzing the organization variable using a histogram, we observed that only a small number of organizations appear in the dataset. Some examples include Gap Inc. and Meta Platforms, which represent exceptions rather than the norm. The organization variable is not representative for the majority of billionaires, and the large number of missing values suggests that wealth is often personal or distributed across multiple ventures, rather than being tied to a single corporation.

Moreover, from Figure 3.10, while analyzing the status variable using a histogram, we observed that most billionaires are self-made, while a smaller portion have inherited wealth. This indicates that a large share of global wealth today is generated through entrepreneurship and personal wealth creation rather than inheritance.

Other categories, such as E, N, R, or Split family fortune, appear to represent special cases of shared or partially inherited family wealth. However, without clear definitions of these categories, it is difficult to analyze them precisely. Overall, this variable suggests that while self-made wealth dominates, inherited wealth still plays an important role in the distribution of billionaire fortunes.

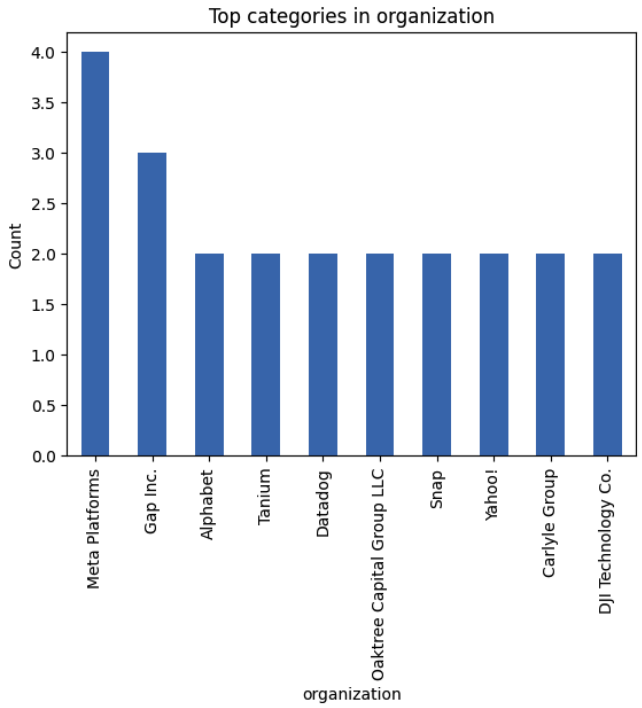


Figure 3.9. Histogram showing the distribution of organizations associated with billionaires.

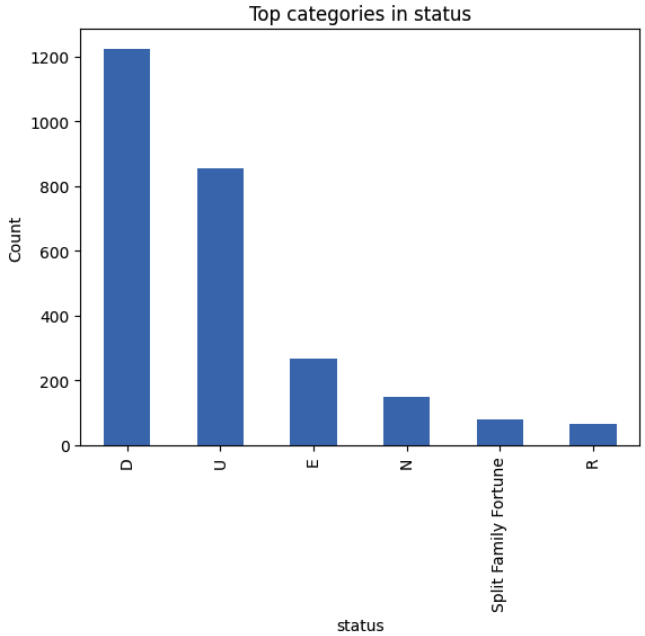


Figure 3.10. Histogram showing the distribution of billionaire status categories (self-made vs inherited wealth).

In addition, from Figure 3.11, while analyzing the gender variable using a histogram, we observed a significant gender gap among billionaires, with more than 2,300 men compared to around 300 women. This disparity may reflect historical and social factors, such as unequal access to education, financing, and business opportunities, rather than differences in ability. Overall, the distribution highlights the presence of persistent gender inequality in the accumulation of global wealth.

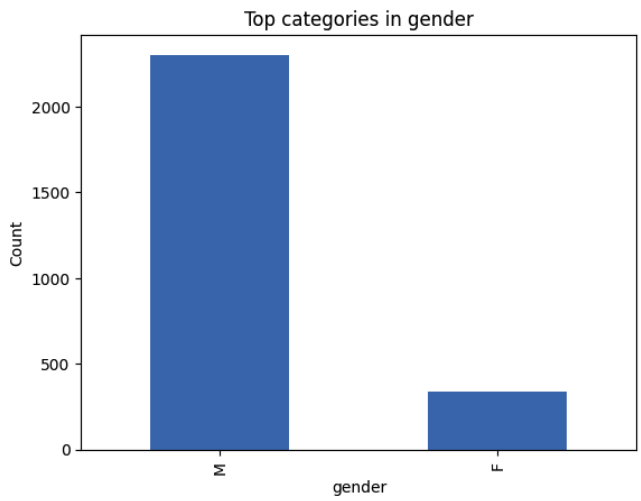


Figure 3.11. Histogram showing the distribution of billionaires by gender.

Additionally, from Figure 3.12, while analyzing the birthDate variable using a histogram, we observed that most billionaires were born during the 1940s, 1950s, and 1960s, suggesting that reaching extreme wealth generally requires time, experience, and many years of building businesses or investments.

The birth years were grouped by decade because the month and day information was not fully reliable. Many records showed birthdays as January 1st, which likely indicates that missing dates were filled with a default placeholder value. Grouping the data by decade allows for a more robust analysis of generational trends among billionaires, avoiding distortions caused by these placeholder dates.

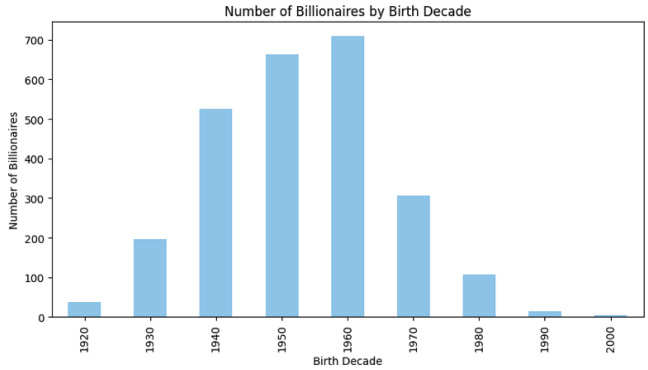


Figure 3.12. Histogram showing the distribution of billionaire birth years grouped by decade.

Also, from Figure 3.13, while analyzing the lastName variable using a histogram, we observed that the most common surnames are Li, Wang, and other Chinese or Asian last names. This pattern is likely related to the large population of China and Asia, where these surnames are extremely common. Therefore, the distribution of last names among billionaires likely reflects global demographic patterns rather than economic or wealth-related factors.

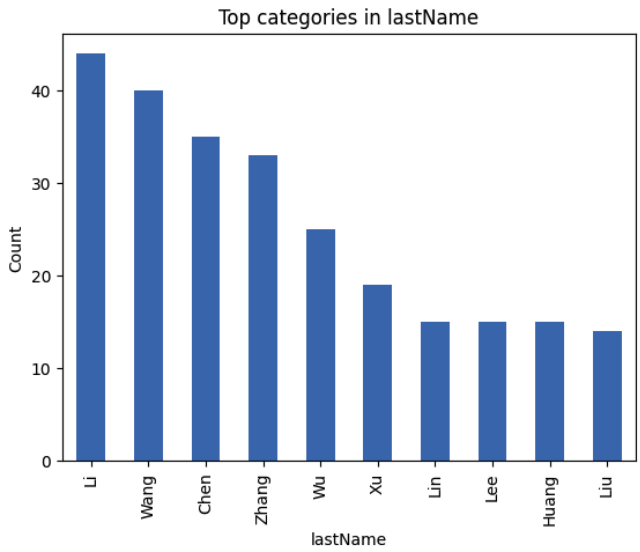


Figure 3.13. Histogram showing the distribution of the most common last names among billionaires.

Additionally, from Figure 3.14, while analyzing the firstName variable using a histogram, we observed that the most common names, such as John, David, Michael, and Richard, are predominantly Western names. This pattern reflects the fact that a large proportion of billionaires reside in Western countries, particularly in the United States and Europe. While last names tend to reflect global population demographics, first names appear to indicate the regional concentration of billionaires in Western economies.

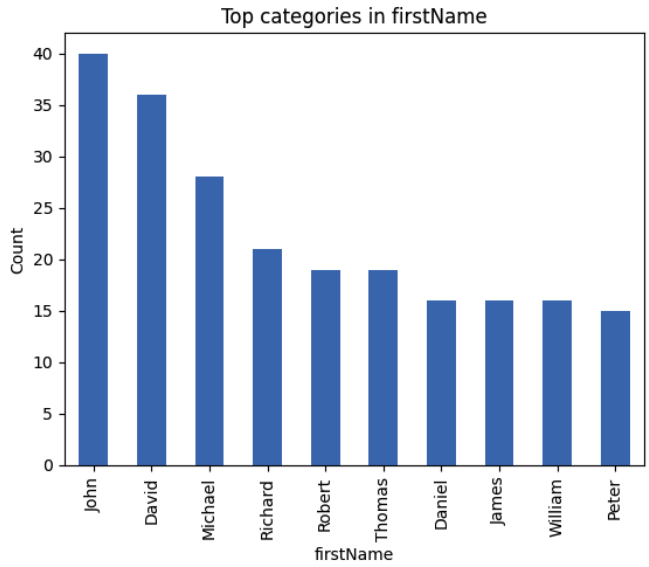


Figure 3.14. Histogram showing the distribution of the most common first names among billionaires.

Furthermore, from Figure 3.15, while analyzing the title variable using a histogram, we observed a few occurrences such as Investor, Founder, and CEO. However, the variable contains 2,330 missing values out of 2,640 records, indicating that the dataset is largely incomplete for this attribute. Due to the very high number

of missing entries, no meaningful analysis can be conducted on this variable.

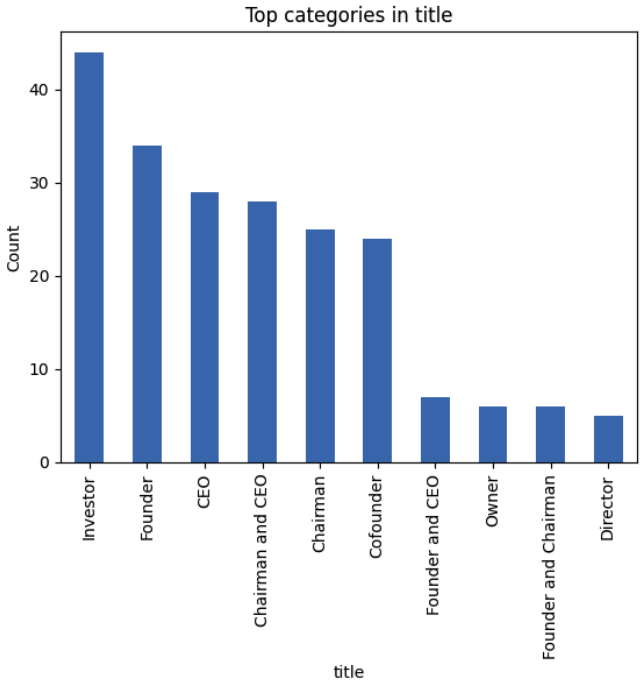


Figure 3.15. Histogram showing the distribution of titles associated with billionaires.

Moreover, from Figure 3.16, while analyzing the state variable using a histogram, we observed that this variable only applies to billionaires located in the United States. The distribution shows a strong concentration of wealth in a few states, with California (175), New York (125), Florida (90), and Texas (65) having the highest number of billionaires, while most other states have fewer than 20. This pattern reflects the presence of major economic, technological, and financial hubs concentrated in a limited number of regions.

Finally, from Figure 3.17, while analyzing the residenceStateRegion variable using a histogram, we observed that this variable groups U.S. states into broader geographic regions. The distribution shows that most billionaires reside in the West (250) and the South (240), followed by the Northeast (190), while the Midwest (50) and U.S. territories (1)contain significantly fewer billionaires. This pattern suggests that wealth in the United States is largely concentrated in major economic and technological hubs located in specific regions.

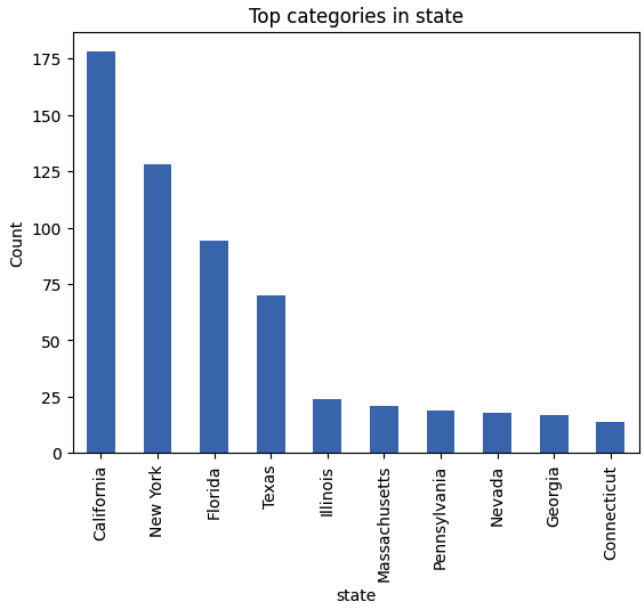


Figure 3.16. Histogram showing the distribution of billionaires across U.S. states.

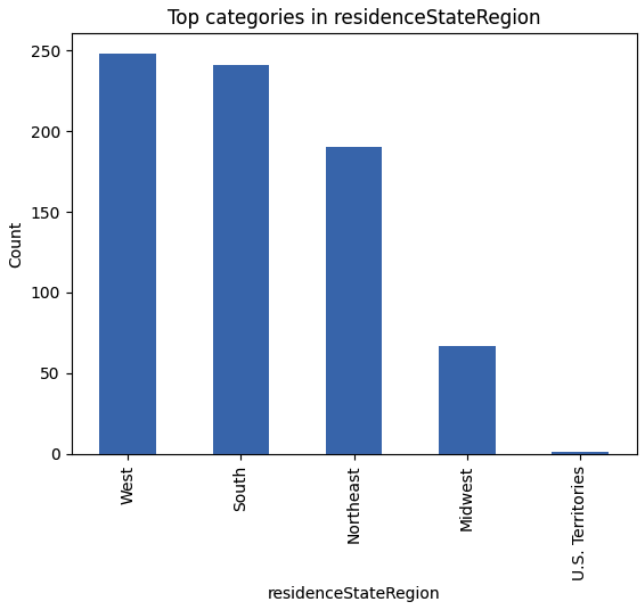


Figure 3.17. Histogram showing the distribution of billionaires across U.S. residence regions.

C) Analysis of numerical variables

Next, we proceed with the analysis of the numerical variables in the dataset. To understand their distributions and identify potential patterns or outliers, we primarily use boxplots, which are useful for visualizing dispersion, central tendency, and extreme values. Additionally, histograms are used in some cases to observe the overall distribution of values. This analysis includes variables related to wealth, age, birth information, and several country-level economic indicators, allowing us to explore both individual and macroeconomic characteristics associated with billionaires.

First, from Figure 3.18, while analyzing the finalWorth variable using a histogram of the log-transformed values, we observed a positively skewed (right-skewed) distribution. Most observations are concentrated between 3.0 and 3.5 on the $\log_{10}$ scale, while the frequency gradually decreases as the values increase. A long tail extends to the right up to approximately 5.2, indicating that a small number of individuals possess significantly higher wealth compared to the majority of the dataset. The logarithmic transformation helps make the distribution easier to visualize by reducing the effect of extreme values.

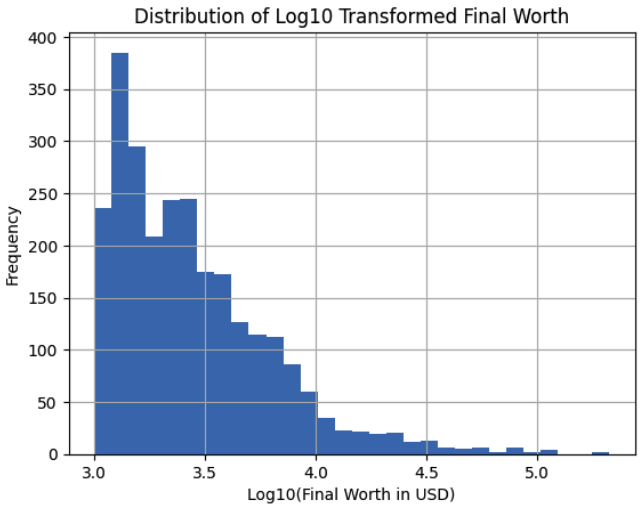


Figure 3.18. Histogram of the $\log_{10}$-transformed final worth distribution among billionaires.

Next, from Figures 3.19 and 3.20, while analyzing the age variable using both a histogram and a boxplot, we observed that the mean age of billionaires is 65.14 years, indicating that most individuals in this group are in their mid-60s. This supports the idea that accumulating extreme wealth generally requires many years of professional experience, entrepreneurship, and long-term investments. The standard deviation of 13.26 shows a moderate dispersion around the mean, meaning that although many billionaires are concentrated near this age, there is still a considerable range. The minimum age is 18 years, suggesting the presence of a few very young billionaires, often associated with technology startups or inherited wealth, while the maximum age reaches 101 years, indicating that some individuals maintain large fortunes well into old age. The quartiles show that 50% of billionaires are between 56 and 75 years old, reinforcing that most belong to middle-aged or older groups. The mode of 60 years is close to both the mean and the median (65), suggesting a relatively balanced distribution without extreme skewness. Additionally, the robust mean (trimmed 10%) is 65.77, which is slightly higher than the standard mean, indicating that extreme ages slightly lower the average but do not significantly alter the overall pattern. Overall, the distribution suggests that extreme wealth is typically accumulated over decades, with a peak around the 60s, while very young or very old billionaires represent rare exceptions.

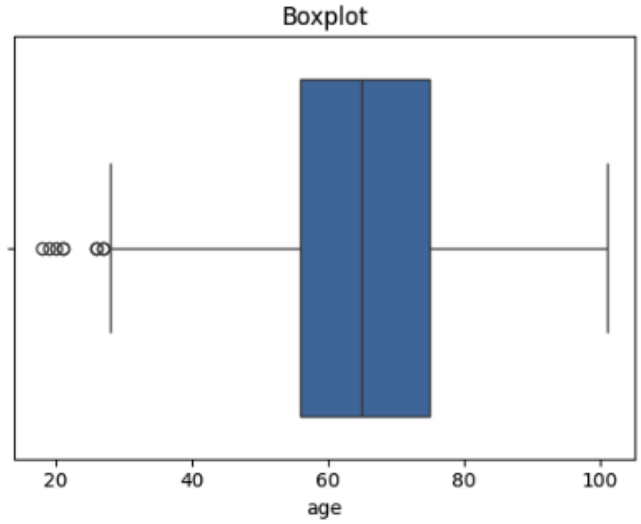


Figure 3.19. boxplot showing the distribution of billionaire ages.

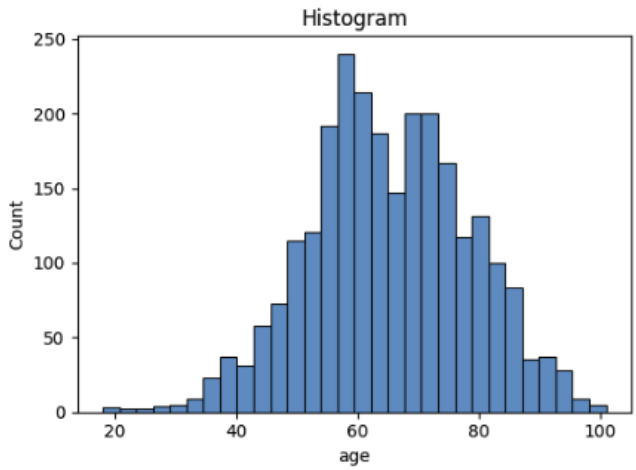


Figure 3.20. Histogram showing the distribution of billionaire ages.

D) Analysis of variables per country

We proceed with the analysis of country-level variables. These variables provide insights into the macroeconomic and demographic context in which billionaires accumulate their wealth. By examining indicators such as GDP, population, life expectancy, tax rates, and education enrollment, we can explore how national economic conditions and social factors relate to the distribution and characteristics of billionaires. This analysis primarily uses histograms and boxplots to highlight trends, variability, and potential outliers among different countries.

First of all, from Figure 3.21, while analyzing the CPI (Consumer Price Index) variable per country using both a histogram and a boxplot, we observed that the mean CPI is 127.76, indicating that, on average, countries in this dataset present moderate CPI levels. This suggests that most countries have similar consumer price environments, without extreme variations for the majority. The standard deviation of 26.45 shows a moderate dispersion around the mean, meaning that although most countries have CPI values clustered around the central range, some variability exists. The minimum CPI is 99.55, reflecting countries with relatively lower purchasing power or economic performance, while the maximum CPI reaches 288.57, highlighting a few

extreme cases that stand out from the general trend. The quartiles show that 50% of countries have CPI values between 117.24 and 125.08, indicating that the majority of countries are concentrated in the lower-middle range. Additionally, the robust mean (trimmed 10%) is 127.09, slightly lower than the standard mean, which suggests that extreme high CPI values moderately affect the average but do not drastically change the overall pattern. Overall, the distribution indicates that while most countries have moderate CPI levels, several countries such as Argentina, Egypt, Nigeria, Turkey, Ukraine, and Uruguay present exceptionally high CPI values, which should be carefully considered in further analyses involving comparisons with other economic indicators.

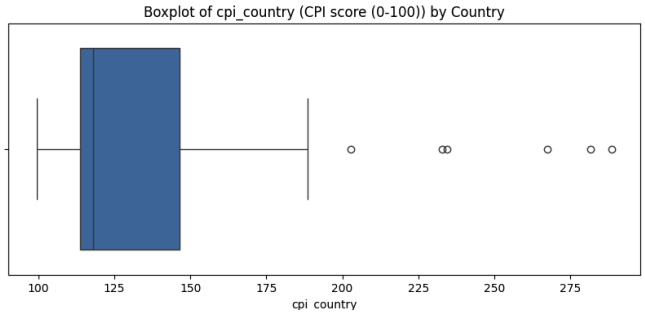


Figure 3.21. Distribution of Consumer Price Index (CPI) per Country.

Next, **from Figure 4.22,** while analyzing the CPI change variable using both descriptive statistics and visualizations, we observed that the mean CPI change is 4.36%, indicating that on average, countries are experiencing moderate inflation. The standard deviation of 3.62 shows some variability between countries, meaning that although most countries have moderate inflation rates, a few exhibit much higher values. The minimum CPI change is -1.9%, reflecting slight deflation in some countries, while the maximum reaches 53.5%, a clear outlier corresponding to extremely high inflation in Argentina. Quartiles reveal that 50% of countries have CPI changes between 1.7% and 7.5%, suggesting that most countries experience low to moderate inflation. The mode of 7.5% aligns with the upper quartile, indicating that moderate inflation is common, though lower and higher changes also occur. Additionally, the robust mean (trimmed 10%) is 4.56%, slightly higher than the standard mean, showing that extreme values, particularly Argentina's very high inflation, moderately affect the average but do not drastically change the overall pattern. Outliers include Argentina (53.5%), Egypt (9.2%), India (7.7%), Nigeria (11.4%), Turkey (15.2%), Ukraine (7.9%), and Uruguay (7.9%), representing countries with significantly higher inflation than the majority. Overall, while most countries experience moderate CPI changes clustering around 1.7–7.5%, a few extreme cases, especially Argentina, could distort averages and economic analysis. Understanding CPI change is important for assessing the economic environment in which billionaires operate, as high inflation can erode real wealth, while stable inflation favors predictable economic growth.

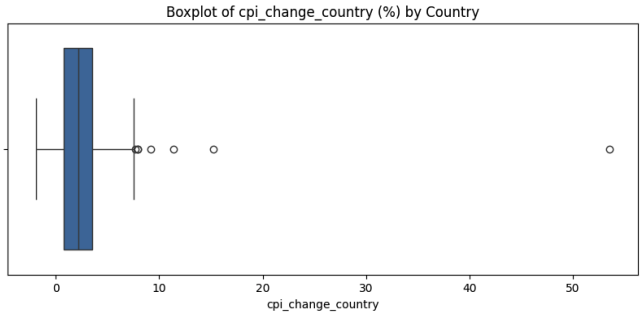


Figure 3.22. Boxplot of CPI Change per Country

Now, from Figure 4.23 , while analyzing the GDP variable using a boxplot, we observed that the mean GDP is approximately 1.16×10^{13}, indicating that the average economic size among countries in the dataset is large, but there is high variability. The standard deviation of 9.58×10^{12} shows that GDP values vary significantly across countries. The minimum GDP is 3.15×10^{9}, representing small economies, while the maximum GDP reaches 2.14×10^{13}, corresponding to very large economies such as the United States and China. Quartiles reveal that 50% of countries have GDPs between 1.74×10^{12} and 2.14×10^{13}, but extreme high values push the median upward. Additionally, the robust mean (trimmed 10%) is approximately 1.18×10^{13}, slightly lower than the raw mean, indicating that extreme high GDPs moderately inflate the average. Outliers include France (2.72×10^{12}), United States (2.14×10^{13}), India (2.61×10^{12}), China (1.99×10^{13}), Germany (3.85×10^{12}), Japan (5.08×10^{12}), United Kingdom (2.83×10^{12}), Italy (2.00×10^{12}), Brazil (1.84×10^{12}), and South Korea (2.03×10^{12}), representing the largest economies in the world. Overall, while most countries have moderate GDP values, a few very large economies heavily influence the mean. Understanding GDP outliers is crucial when analyzing billionaire wealth in relation to the size of the economy, as most billionaires are likely concentrated in these high-GDP countries.

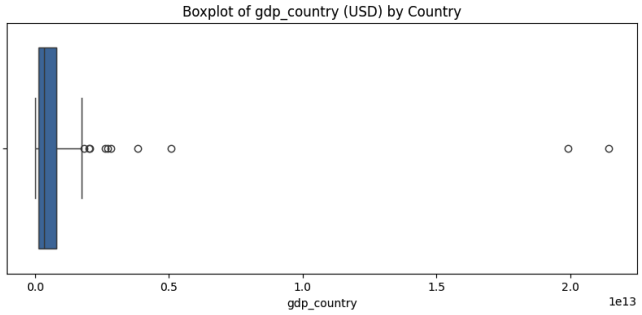


Figure 3.23. Boxplot of GDP per Country.

Aditionally, from Figure 3.24, while analyzing the gross tertiary education enrollment variable using a boxplot, we observed that the mean enrollment rate is approximately 67.23%, indicating that on average, around two-thirds of the population at tertiary education age (or outside it) are enrolled in higher education. The standard deviation of 21.34 shows moderate variability between countries, meaning that some countries have very low enrollment while others reach extremely high rates. The minimum enrollment is 4.0%, reflecting very limited access to tertiary education in some countries, whereas the maximum is 136.6%, an outlier due to the "gross" nature of the indicator, which can exceed 100% when adults outside the official age group are enrolled. Quartiles reveal that 50% of countries have enrollment

rates between 50.6% and 88.2%. The robust mean (trimmed 10%) is approximately 68.75%, slightly higher than the simple mean, indicating that extremely low enrollments slightly pull the average down. No outliers were detected using the IQR method, showing that while there are countries with low or high enrollment, none are extreme compared to the rest. Overall, most countries have moderate to high tertiary education enrollment rates, typically between 50% and 88%, reflecting generally good access to higher education and highlighting human capital development and potential for innovation and wealth creation.

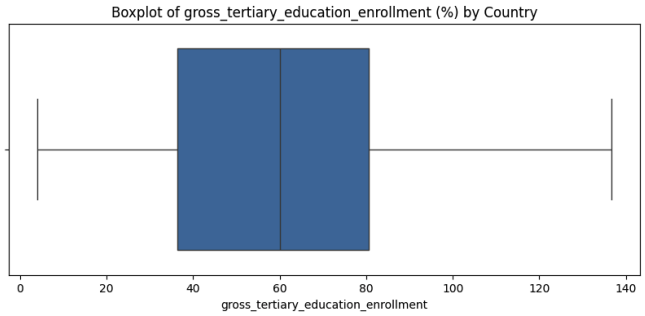


Figure 3.24. Boxplot of Gross Tertiary Education Enrollment per Country.

Next, from Figure 3.25, while analyzing the gross primary education enrollment variable using a boxplot, we observed that the mean enrollment rate is approximately 102.86%, slightly above 100% due to the "gross" nature of the indicator, which includes students outside the official primary school age. The standard deviation of 4.71 shows low variability between countries, indicating that most countries achieve very similar primary enrollment rates. The minimum enrollment is 84.7%, reflecting a few countries with lower coverage, while the maximum reaches 142.1%, an extreme case likely caused by over-aged or under-aged students being enrolled. Quartiles reveal that 50% of countries have enrollment rates between 100.2% and 102.6%, and the mode of 101.8% aligns closely with both the median and the upper quartile, confirming the overall stability of primary education enrollment. The robust mean (trimmed 10%) is approximately 101.92%, slightly higher than the simple mean, showing that extreme values have minimal effect on the average. Outliers include Sweden (126.6%), Nigeria (84.7%), Romania (85.2%), and Nepal (142.1%), representing countries with unusually high or low primary enrollment rates, mostly due to enrollment outside the official age range or incomplete coverage. Overall, most countries have very high primary education enrollment, typically around 100–102%, reflecting near-universal access to primary schooling globally, with a very tight distribution compared to other indicators such as GDP or CPI.

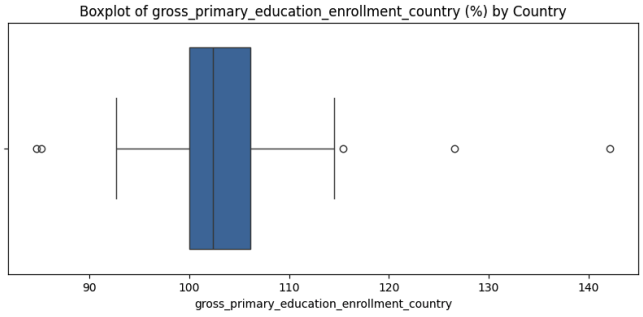


Figure 3.25. Boxplot of Gross Primary Education Enrollment per Country.

Next, from Figure 3.26, while analyzing the life expectancy variable using a boxplot, we observed that the mean life expectancy is approximately 78.12 years, indicating that on average, people live almost 78 years across the countries in this dataset. The standard deviation of 3.73 shows moderate variability, meaning that most countries cluster around the mid to high 70s. The minimum life expectancy is 54.3 years, reflecting countries with significantly lower health outcomes, while the maximum reaches 84.2 years, indicating countries with very high longevity. Quartiles reveal that 50% of countries have life expectancy between 77.0 and 80.9 years, and the mode of 78.5 years aligns with the median, showing that many countries fall close to the middle of the distribution. The robust mean (trimmed 10%) is approximately 78.49 years, very close to the simple mean, indicating that extreme low or high values have minimal effect on the average. Outliers include Nigeria (54.3 years), South Africa (63.9 years), and Tanzania (65.0 years), representing countries with unusually low life expectancy likely due to lower healthcare access, higher disease burden, or socio-economic challenges. Overall, most countries have life expectancy between 77 and 81 years, reflecting relatively high global longevity, with a few outliers highlighting regions facing significant public health challenges.

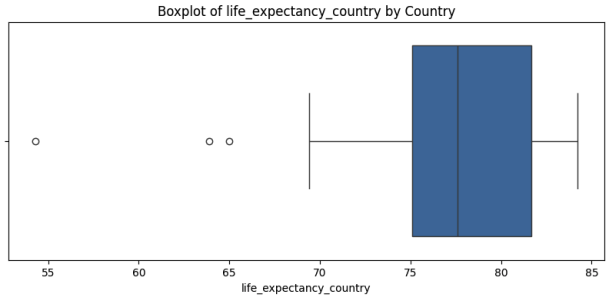


Figure 3.26. Boxplot of Life Expectancy per Country.

Furthermore, from Figure 3.27 ,while analyzing the tax revenue variable using a boxplot, we observed that the mean tax revenue is approximately 12.55%, indicating that, on average, countries collect about 12–13% of their GDP in taxes. The standard deviation of 5.37 shows moderate variability between countries, meaning that some countries collect significantly more or less depending on their economic structure and fiscal policies. The minimum tax revenue is 0.1%, reflecting extremely low taxation in countries like the United Arab Emirates, while the maximum reaches 37.2%, as in Algeria, representing high fiscal pressure relative to GDP. Quartiles reveal that 50% of countries collect between 9.6% and 12.8% of GDP in taxes, and the mode of 9.6% aligns with the 25th percentile and median, indicating that many countries cluster around this level of taxation. The robust mean (trimmed 10%) is 11.42%, slightly lower than the raw mean, showing that countries with very high tax revenue moderately elevate the average. No outliers were detected using the IQR method, though values like the UAE's 0.1% are exceptionally low in reality. Overall, most countries have moderate tax revenue relative to their GDP, typically between 9.6% and 12.8%, with high values reflecting strong fiscal systems and low values reflecting alternative economic models.

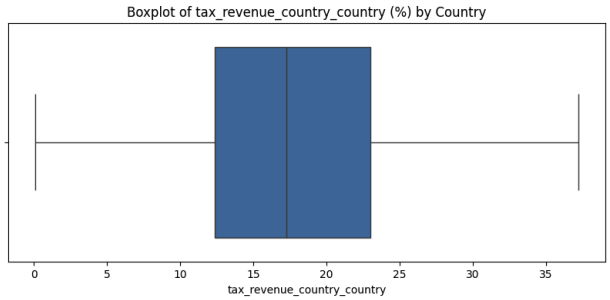


Figure 3.27. Boxplot of Tax Revenue per Country.

Also, from Figure 3.28, while analyzing the total tax rate variable using a boxplot, we observed that the mean total tax rate is approximately 43.96%, indicating that, on average, countries impose a moderate to high tax burden on businesses and individuals. The standard deviation of 12.15% shows considerable variability between countries, meaning that some have much lower or higher rates depending on their fiscal system. The minimum total tax rate is 9.9%, reflecting countries with very low tax burdens, such as tax havens or nations with low taxation policies, while the maximum reaches 106.3% in Argentina, an extreme case where effective combined taxes exceed 100%, reflecting very high income, corporate, and social contributions. Quartiles reveal that 50% of countries have total tax rates between 36.6% and 59.1%, and the mode of 36.6% aligns with the first quartile, showing that many countries have relatively lower rates within the typical range. The robust mean (trimmed 10%) is 44.22%, slightly higher than the simple mean, indicating that extreme values do not overly skew the average. Only one outlier was detected: Argentina (106.3%), representing an exceptionally heavy combined tax burden. Overall, most countries have total tax rates between 36% and 59%, with low extreme values reflecting light taxation policies and high extremes like Argentina showing extraordinarily high effective tax burdens. Compared to tax revenue, this variable reflects the effective tax burden rather than total revenue collected.

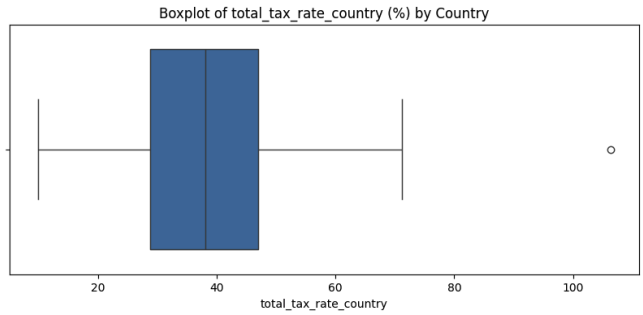


Figure 3.28. Boxplot of Tax Revenue per Country.

Next, from Figure 3.29, while analyzing the population variable using a boxplot, we observed that the mean population is approximately 510 million, heavily influenced by very populous countries such as China, India, and the United States. The standard deviation of 554 million shows extremely high variability, reflecting populations ranging from very small countries to giants like China and India. The minimum population is 38,019, representing very small states or territories, while the maximum reaches 1,397,715,000 in China, the most populous country. Quartiles reveal that 50% of countries have populations between 328.2 million and 1.366 billion, but the distribution is strongly skewed due to a few extremely large countries.. The robust mean (trimmed 10%) is 462 million, slightly lower than the raw mean, showing that trimming extreme values reduces the influence of the largest countries. Outliers include China (1.397 billion), India (1.366 billion), United States (328 million), Indonesia (270 million), Brazil (212 million), and Nigeria (201 million), representing countries with exceptionally large populations. Overall, population distribution among countries is highly skewed, with a small number of very populous countries driving up the mean, making the median and robust mean better indicators of a typical country's population. This is important when analyzing metrics like GDP, tax revenue, or total tax rate, as large countries can dominate totals and averages, potentially masking trends in smaller countries.

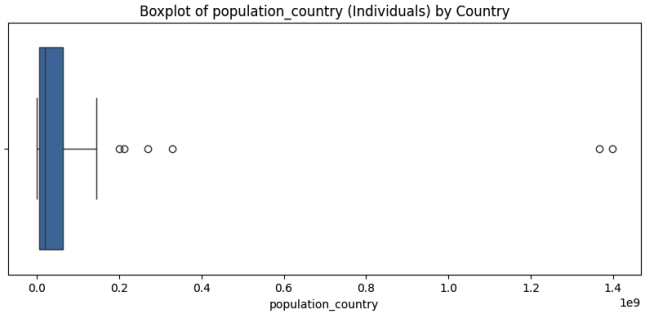


Figure 3.29. Boxplot of Population per Country.

Ultimately, The correlation matrix (See figure 3.30) of country-level variables reveals several noteworthy relationships. First, GDP shows a strong positive correlation with population, indicating that countries with larger populations tend to have higher total economic output, which aligns with expectations. Additionally, GDP is moderately positively correlated with tertiary education enrollment and life expectancy, suggesting that wealthier countries tend to invest more in higher education and achieve better health outcomes. Interestingly, CPI and CPI change show weak correlations with most economic variables, indicating that inflation levels are largely independent of country size, GDP, or education indicators. Tax-related variables present some expected patterns: total tax rate is moderately positively correlated with tax revenue, reflecting that higher effective taxation generally results in higher government revenue relative to GDP. Overall, the matrix highlights the expected socio-economic links—population and GDP, GDP with human capital and health—while variables like CPI remain mostly uncorrelated with these measures, emphasizing their independent dynamics in the dataset.

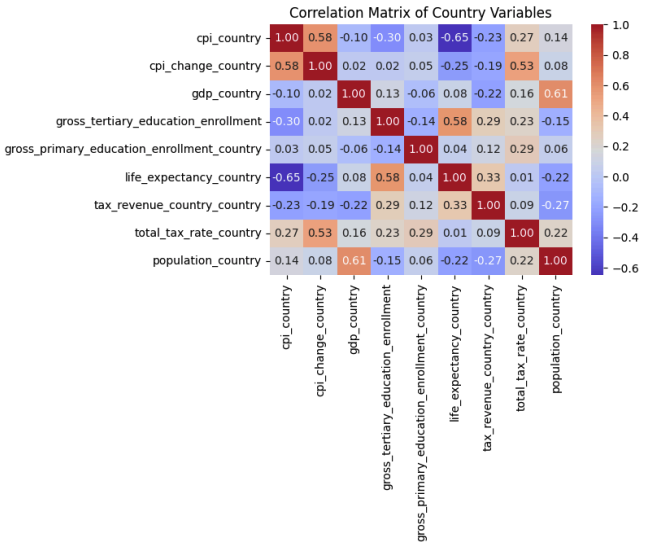


Figure 3.30 . Correlation Matrix of Country-Level Variables.

Exploratory Analysis for the Classification Task

The purpose of the exploratory analysis in this phase is to identify structural patterns that may allow differentiation between national wealth ecosystems. Since the classification task aims to group countries into conceptually distinct categories — *Innovation Ecosystem*, *Rentier Paradise*, and *Extraction Economy* — the analysis focuses on three main structural dimensions: wealth composition, sectoral structure, and demographic characteristics.

Distribution of Billionaires by Country

The distribution of billionaires by country is shown in Figure 3. 31. The chart highlights a strong geographic concentration of extreme wealth in a limited number of countries. The **United States** clearly dominates with the highest number of billionaires, followed by **China**, while **India** ranks third at a considerable distance. Other countries such as **Germany**, **United Kingdom**, **Russia**, **France**, **Hong Kong**, **Italy**, and **Singapore** present significantly lower values.

This distribution suggests a clear asymmetry in the global concentration of wealth. Although this pattern may partly reflect the economic size of these countries, it may also indicate structural differences in national systems of wealth creation and capital accumulation. From an analytical perspective, this variable helps identify countries with sufficient representation for structural analysis and provides an initial signal of economic scale and capital concentration. However, the skewness observed in the distribution indicates that additional structural variables are necessary to meaningfully differentiate wealth ecosystems.

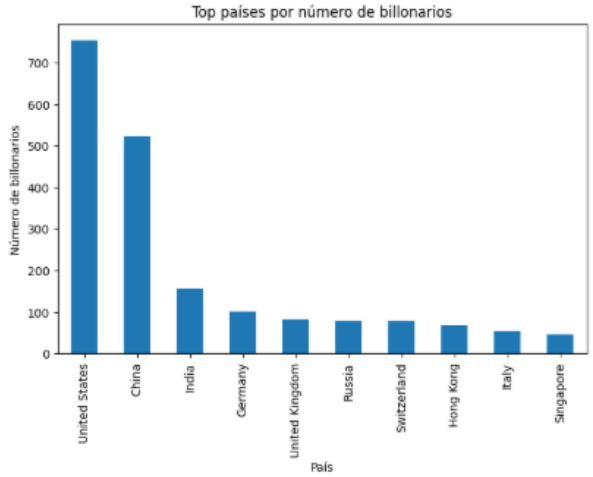


Figure 3.31. Histogram of Distribution of billionaires by country.

Proportion of Self-Made Wealth

The proportion of self-made billionaires across countries is illustrated in Figure 3.32. The chart reveals substantial cross-country variation in the origin of extreme wealth. Some countries show very high proportions of self-made billionaires, indicating that most wealth is generated through entrepreneurial activity and new ventures. In contrast, other countries display lower values, suggesting a stronger role of inherited wealth and established family fortunes.

This variability highlights important structural differences in how wealth is created and transmitted across national economies. Countries with higher self-made ratios may reflect more dynamic, innovation-driven environments that favor entrepreneurship and new capital formation. Conversely, lower proportions may indicate economic systems where wealth is more strongly linked to inheritance or long-standing capital ownership.

From a classification perspective, this variable is particularly informative because it helps distinguish between different models of wealth generation. Specifically, it allows the analysis to differentiate between economies characterized by **dynamic entrepreneurial wealth creation** and those where **intergenerational capital reproduction** plays a more dominant role. The heterogeneity observed in the figure therefore supports the hypothesis that structurally distinct national wealth ecosystems exist.

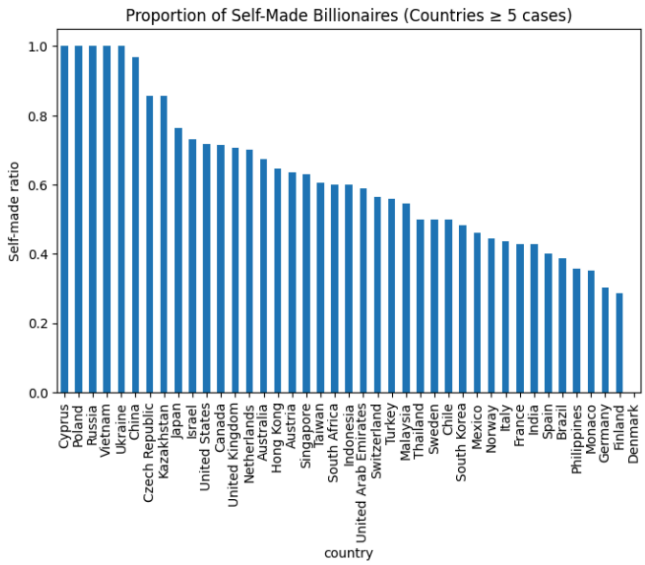


Figure 3.32. Histogram of Proportion of self-made billionaires by country (countries with at least five cases).

Sectoral Distribution of Wealth

The sectoral distribution of billionaire wealth is presented in Figure 3.33. The chart shows the industries with the highest number of billionaires, highlighting how extreme wealth is concentrated in specific economic sectors. **Finance & Investments** appears as the leading sector, followed by **Manufacturing** and **Technology**, which also account for a large share of billionaires. Other important sectors include **Fashion & Retail**, **Food & Beverage**, **Healthcare**, and **Real Estate**, while industries such as **Energy** and **Media & Entertainment** show comparatively lower representation.

This distribution suggests that billionaire wealth is not evenly spread across industries but tends to concentrate in sectors with strong capital accumulation potential or high market scalability. For example, technology-driven sectors may reflect innovation-based economic systems, while finance and real estate may indicate environments characterized by strong capital markets or asset-based wealth accumulation.

From an analytical perspective, sectoral distribution is particularly relevant for classification because it helps identify the main drivers of wealth generation within national economies. It also allows the distinction between more diversified economic structures and those that are highly specialized in specific industries. These sectoral patterns therefore represent a key structural dimension for understanding and classifying different national wealth ecosystems.

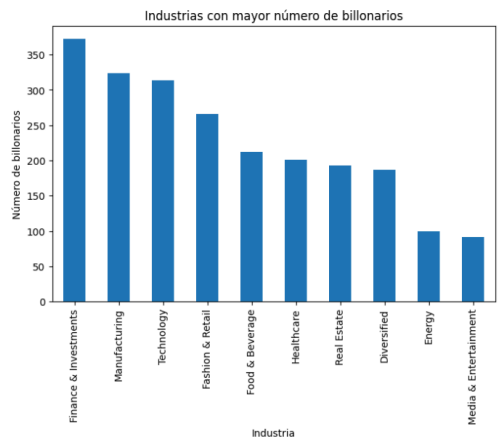


Figure 3.33. Histogram of Sectoral distribution of billionaires.

Demographic Structure

The age distribution of billionaires is illustrated in **Figure 33**. The histogram shows that most billionaires fall within the age range of approximately **50 to 75 years**, with the highest concentration around the early sixties. Very young billionaires are relatively rare, and the frequency gradually decreases at older ages, although individuals above 80 are still present.

This pattern suggests that the accumulation of extreme wealth typically occurs over long time horizons, often after decades of business activity, investment, or capital growth. The distribution therefore reflects the time required to build large-scale economic success and consolidate financial influence.

From an analytical perspective, age distribution can be interpreted as a proxy for **generational renewal within national wealth systems**. A lower average age could indicate faster entrepreneurial turnover and more dynamic economic environments, while a higher concentration in older age groups may suggest more stable or inheritance-based wealth structures.

Although this demographic dimension is not decisive on its own, it contributes to a broader structural understanding of the **maturity and renewal dynamics of wealth ecosystems**, complementing other economic and sectoral indicators used in the analysis.

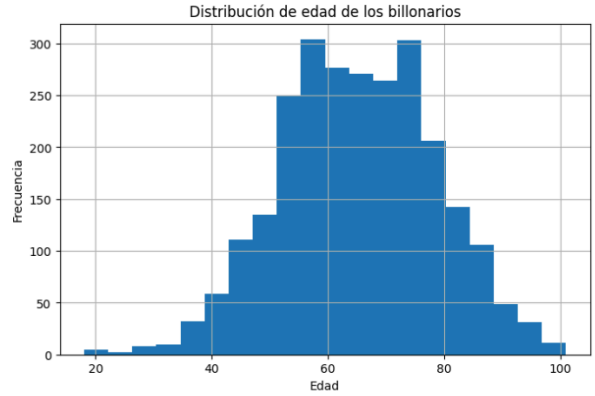


Figure 3.34. Age distribution of billionaires.

Education and Innovation Ecosystems

The relationship between tertiary education enrollment and the proportion of self-made wealth is illustrated in **Figure 34**. The scatter plot explores whether higher levels of human capital formation are associated with greater entrepreneurial wealth creation. A preliminary visual inspection suggests that countries with higher tertiary education enrollment often display higher proportions of self-made billionaires, although the relationship is not perfectly linear and shows considerable variability across observations.

This pattern may indicate a potential connection between **human capital development and entrepreneurial wealth creation**. Countries with stronger education systems may provide more favorable conditions for innovation, business creation, and technological development, which in turn can facilitate the emergence of self-made wealth.

From a conceptual perspective, this relationship supports the idea of an **innovation-driven ecosystem**, where knowledge-based industries and highly educated workforces play a central role in wealth generation. However, the dispersion of points also suggests that education alone is not sufficient to explain the formation of entrepreneurial wealth and that additional institutional and economic factors are likely involved.

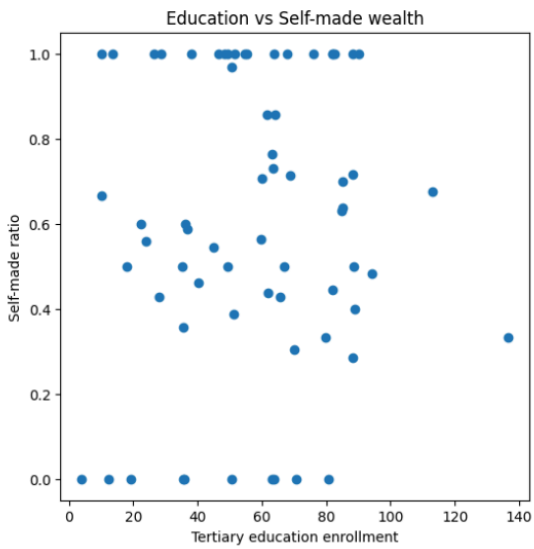


Figure 3.35. The scatter plot of Relationship between tertiary education enrollment and self-made wealth ratio.

IV. Data Preprocessing

Data preprocessing is a crucial phase in the data mining process, focused on transforming raw data into a clean and structured format suitable for analysis. This stage involves tasks such as handling missing values, reducing redundancy, standardizing variables, and eliminating irrelevant features. Proper preprocessing ensures data quality, improves interpretability, and enhances the performance of subsequent data mining techniques.

A. Level 1 - Base Data Cleaning

1) Check for duplicates

During the data preprocessing stage, duplicate records were identified based on the attribute personName, which serves as the primary identifier for individuals in the dataset. The presence of duplicate entries may introduce bias and inconsistencies in subsequent analyses, particularly when aggregating financial or demographic attributes.

To address this issue, a deduplication procedure was applied in which all records sharing the same value in personName were considered duplicates. For each group of duplicated entries, only the first occurrence was retained, while all subsequent records were removed. This strategy ensures that each individual is represented by a single observation in the dataset, preserving data integrity while minimizing information loss.

After the removal process, the dataset dimensions were updated, and a validation check confirmed the absence of remaining duplicates in the personName field.

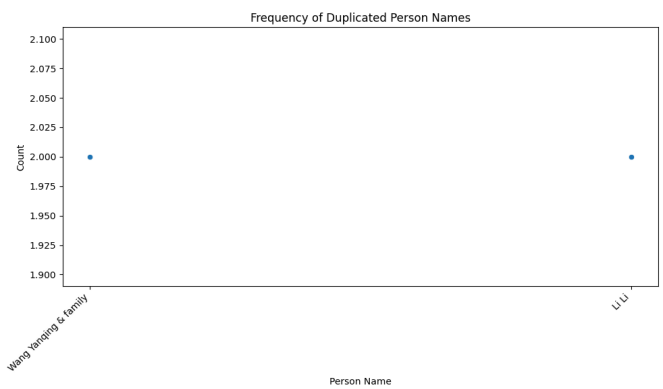


Figure 4.1. The scatter plot of the frequency of duplicated person names

2) Handling Missing Values

2.1. Imputation of Missing Values in the *country* Column

The dataset contained missing values in the categorical attribute country, which represents the country of residence of each individual. Specifically, 38 entries (~1.44% of the dataset) were identified as null values in this field. Given the importance of preserving all observations for downstream analysis, row deletion was not considered an appropriate strategy. Instead, a structured imputation approach was applied to estimate missing values while maintaining data integrity and minimizing information loss.

Following the principle of hierarchical recovery of missing categorical attributes, the missing values were inferred using the most contextually reliable information available within each record.

Hierarchical Imputation Strategy

A four-level hierarchical imputation procedure was implemented:

- countryOfCitizenship (Primary Source):
  Missing country values were first filled using the countryOfCitizenship attribute, as citizenship is strongly correlated with residence and is considered the most reliable available proxy.
- City-Based Inference:
  If both country and countryOfCitizenship were missing or insufficient, the city field was used. A city-to-country mapping, constructed from existing dataset distributions, was applied to infer the corresponding country.
- State/Region-Based Inference:
  For records containing a state or residenceStateRegion, the country was inferred as United States, since state-level entries in the dataset predominantly correspond to U.S.-based individuals.
- Default Assignment (Unknown):
  In cases where none of the previous strategies could be applied, the value was assigned as “Unknown”, ensuring row preservation while explicitly marking missing information.

2.2. Imputation of Missing Values in the city Column

The dataset contained missing values in the categorical attribute city, which represents the city of residence of each individual. A total of 72 records (~2.73% of the dataset) were identified with missing values in this field. These missing values included both true null entries (NaN) and previously imputed placeholders labeled as "Unknown", which were treated equivalently as missing information for the purpose of recovery.Given the importance of preserving all observations, a deletion-based approach was discarded. Instead, a hierarchical imputation strategy was applied to infer missing city values using progressively less granular but contextually relevant information.

This procedure follows the principle of contextual estimation for categorical variables, as described by Tan et al. (Section 2.2.1), where missing values are inferred using the most similar available records.

- Mode by Country (Internal Estimation):
  For each record with a missing or "Unknown" city, the most frequent city among individuals sharing the same country was used. This leverages intra-country similarity under the assumption that wealth concentration is often geographically clustered.
- SKnown Billionaire Hub by Country (External Knowledge Base):
  If no dominant city existed within the dataset for a given country, a predefined mapping of globally recognized billionaire hubs per country was used. This mapping was constructed using publicly available wealth distribution reports and Forbes 2023 data.

2.3. Imputation of Missing Values in the age Column

The dataset contained missing values in the numerical attribute age, which represents the current age of each individual. A total of 65 records (~2.46% of the dataset) were identified as having missing values in this field.

Given the objective of preserving all observations for subsequent analysis, a deletion-based approach was not considered appropriate. Instead, an imputation strategy was applied to estimate missing values while maintaining the statistical properties of the distribution.

Proposed and Applied Strategy

- Recalculation from birthDate and date: Instead of relying solely on the provided age column, the age is now explicitly calculated using the birthDate and date columns (the date the data was collected). This provides a more consistent and verifiable age for each billionaire.
- Median Imputation for Remaining Nulls: After recalculating, any age values that might still be null (e.g., if birthDate was also missing or invalid for a record) are filled with the global median of the newly calculated age values. The median is preferred over the mean because the age distribution, while generally central, can still be influenced by outliers (very young or very old billionaires), and the median provides a more robust measure of the typical age.
- Why global median?: Given that a small percentage (less than 5%) of values required imputation after recalculation, the global median provides a sufficiently accurate and simple estimate without the need for more complex group-based imputation, which would offer only marginal gains in precision for such a small fraction of data.

2.4. Elimination of the *organization* Column

The dataset contained the categorical attribute organization, which represents the organization or company associated with each billionaire. A total of 2313 records (~87% of the dataset) were identified as missing values in this field. Given the extremely high proportion of missing data, a deletion-based or imputation-based approach was not considered appropriate. Instead, a feature elimination strategy was applied to ensure data quality and analytical reliability. when a column exhibits an overwhelmingly high proportion of missing values, removing the attribute is the most appropriate preprocessing strategy. In this case, the organization attribute could not be reliably reconstructed due to the absence of sufficient contextual or correlated variables. At 87% missingness, retaining this attribute would introduce significant bias, as any analysis would be based only on a small and non-representative subset of observations. Therefore, the column was considered to contain more noise than informative signal. As a result, the organization column was permanently removed from the dataset, with no records being deleted during this process

2.5. Elimination of the state Column

The dataset contained the categorical attribute state, representing the state of residence of each billionaire. A total of 1885 records (~71% of the dataset) were identified as missing values in this field. Proposed and Applied Strategy: Following

standard data preprocessing principles, attributes with a majority of missing values are considered unsuitable for reliable analysis and are typically removed from the dataset. In this case, the state attribute exhibited an excessively high proportion of missingness (~71%), making it non-representative of the underlying population. From an analytical perspective, the city attribute was selected as the preferred geographic level of granularity. The city field provides more detailed location information and was successfully completed through the previously applied hierarchical imputation strategy. As a result, retaining both city and state would introduce redundancy without adding meaningful analytical value.

The residenceStateRegion attribute was removed following a dual justification based on data completeness and redundancy.

First, the column exhibited a high proportion of missing values (~71%), rendering it unsuitable for reliable analysis. attributes with a majority of missing data should be eliminated, as imputation in such cases would introduce significant uncertainty and bias. Second, the attribute was considered redundant in the context of the existing dataset structure. The city variable, which provides a more granular and complete geographic representation, was already retained and fully imputed through hierarchical recovery methods. Compared to city, the residenceStateRegion attribute offers a broader and less informative geographic resolution, without adding additional analytical value.

2.6. Imputation of Country-Level Socioeconomic, Demographic, and Temporal Attributes

This section describes the treatment of missing values in both country-level indicators and individual temporal attributes within the dataset. The affected variables include *cpi_country, cpi_change_country,gdp_country,gross_tertiary_education_enrollment, life_expectancy_country, tax_revenue_country_country, total_tax_rate_country, population_countr*y, as well as *birthYear* and birthDate. Missing values in the country-level variables ranged between approximately 6.21% and 6.97%, while the temporal attributes exhibited 76 missing entries (~2.88%).

Country-Level Variables

All country-level variables represent macroeconomic, social, or demographic indicators, which are inherently shared across individuals within the same country. These include measures of inflation, economic output, education, health, taxation, and population. Given this structure, the most appropriate imputation strategy is to leverage within-country consistency.

Missing values were estimated using the most contextually similar observations, defined at the country level. A group-based imputation strategy was applied, where missing values were replaced with the median value of the corresponding variable within each country group. The median was selected due to its robustness against outliers and potential inconsistencies in macro-level data.In cases where all values for a given variable within a country were missing, a global median fallback was applied to ensure complete data coverage. This two-step approach preserves both local consistency and global robustness.

As a result, all missing values across country-level variables were successfully imputed. No records were removed, and the dataset size remained unchanged.

3) Delete Irrelevant Columns

The removal of irrelevant and redundant attributes was carried out to simplify the dataset, reduce dimensionality, and improve the efficiency of subsequent data mining processes. This step focuses on identifying variables that either do not provide meaningful information, contain redundant data, or can be derived from other attributes.

First, the variables personName, firstName, and lastName were analyzed. As observed in the categorical analysis, these attributes present a very high number of unique values, since most records correspond to different individuals. Specifically, personName contains approximately 99.92% unique values, while firstName and lastName also exhibit high uniqueness levels (67.05% and 65.76%, respectively).

TABLE 4.1
PERCENTAGE OF UNIQUE VALUES FOR PERSONNAME, FIRSTNAME AND LASTNAME ATTRIBUTES

Attribute	Unique Values Percentage
personName	99.92%
firstName	67.05%
lastName	65.76%

Additionally, personName is a direct composition of firstName and lastName, making these two attributes redundant. Therefore, firstName and lastName were removed, retaining personName as the single identifier of each individual, as it avoids ambiguity and allows results to be easily associated with specific individuals during analysis and interpretation.

Next, the attributes related to birth information (birthDate, birthYear, birthMonth, and birthDay) were evaluated. These variables represent the same information at different levels of granularity.

TABLE 4.2
5 ROWS OF BIRTHDATE, BIRTHYEAR, BIRTHMONTH AND BIRTHDAY ATTRIBUTES

birthDate	birthYear	birthMonth	birthDay
3/5/1949	1949	3	5
6/28/1971	1971	6	28
1/12/1964	1964	1	12
8/17/1944	1944	8	17
8/30/1930	1930	8	30

In particular, birthYear, birthMonth, and birthDay can be directly derived from birthDate, making them redundant. Including multiple representations of the same information does not contribute additional knowledge and may increase data complexity. For this reason, birthMonth and birthDay were removed, while the remaining attributes preserve sufficient information for potential age-related analysis.

The variable rank was also identified as redundant. This attribute is directly derived from finalWorth, as billionaires are ranked according to their net worth.

TABLE 4.3
FIRST 5 SORTED ROWS OF RANKING VS FINALWORTH

rank	finalWorth
1	211000
2	180000
3	114000
4	107000
5	106000

Consequently, rank does not represent an independent source of information but rather an ordered transformation of an existing variable. To avoid redundancy, rank was removed, and finalWorth was retained as it preserves the original quantitative information.

In contrast, the attributes country and countryOfCitizenship were carefully compared to determine whether they represented redundant or complementary information. While country typically refers to the place of residence, countryOfCitizenship represents the nationality of the individual. An analysis of both variables revealed that there are instances where their values differ, indicating that they capture distinct aspects. Therefore, both attributes were retained, as they may provide valuable insights for further analysis.

Finally, the columns latitude_country and longitude_country were removed. These attributes were considered redundant with respect to the country variable, which already provides a more interpretable representation of location. Additionally, since the city attribute was preserved and offers a more precise level of geographic detail, country-level coordinates were deemed unnecessary. Following standard data preprocessing guidelines, these columns were eliminated entirely without removing any records from the dataset.

4) Standardization of Industries And Sources

The standardization of the industries and source attributes was performed to reduce the high cardinality and heterogeneity present in these categorical variables. Initially, both attributes contained a large number of highly specific and granular values, which made it difficult to identify meaningful patterns and increased the complexity of subsequent data mining tasks.

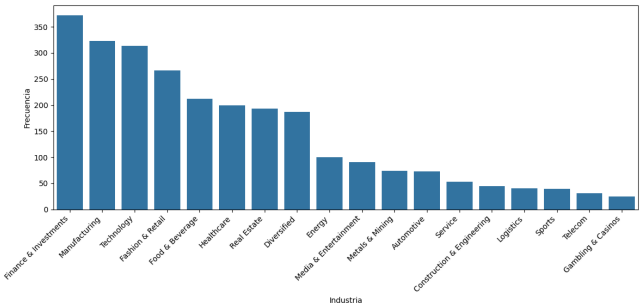


Figure 4.2. Bar plot of industries frequency showing the frequency of each original industry.

To address this issue, an exploratory analysis was conducted by examining the frequency distribution of each variable, allowing the identification of dominant categories and the overall dispersion of values.

For the industries attribute, the original 18 categories were consolidated into 9 broader macro-sectors. This transformation was designed to reduce granularity while preserving the semantic meaning of each category. The grouping follows common economic classification principles, enabling a more interpretable and structured representation of the data.

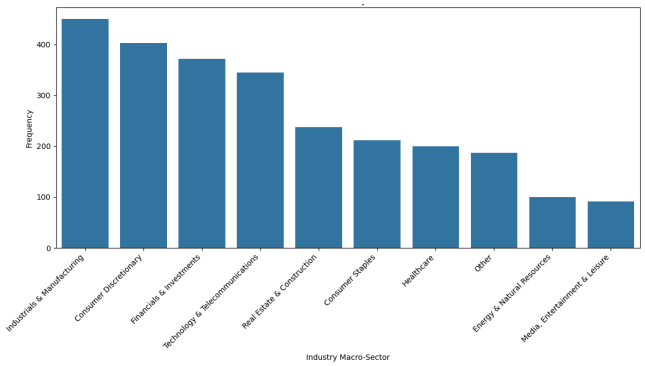


Figure 4.3. Bar plot of industries showing the distribution of the new industry macro-sectors.

As a result, similar industries (such as Technology and Telecommunications, or Real Estate and Construction) were merged into unified macro-sectors, facilitating comparative analysis and improving the robustness of pattern discovery.

A similar approach was applied to the source attribute; however, this variable presented a significantly higher level of cardinality, with hundreds of unique values describing very specific origins of wealth.

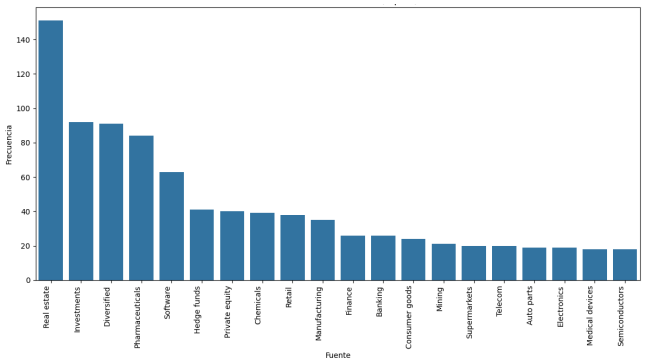


Figure 4.4. Bar plot of source showing the top 20 most frequent sources.

To handle this, a keyword-based mapping strategy was implemented. Each source entry was normalized to lowercase and matched against a predefined set of keywords associated with broader macro-categories. This process allows flexible and scalable classification, as it captures variations in text while assigning each entry to a consistent macro-sector such as Technology, Finance and Investments, or Retail and Consumer Goods.

The mapping process follows a priority-based matching strategy, where categories with more specific or distinctive keywords are evaluated first to reduce ambiguity in classification. If no match is found, the value is assigned to an "Other" category, ensuring completeness of the transformation. This approach significantly reduces dimensionality while maintaining relevant information about the origin of wealth.

Finally, the resulting distributions of both industries and source macro-sectors were analyzed and visualized. The transformation leads to a more balanced and interpretable dataset, enabling clearer insights and improving the performance of downstream analytical methods.

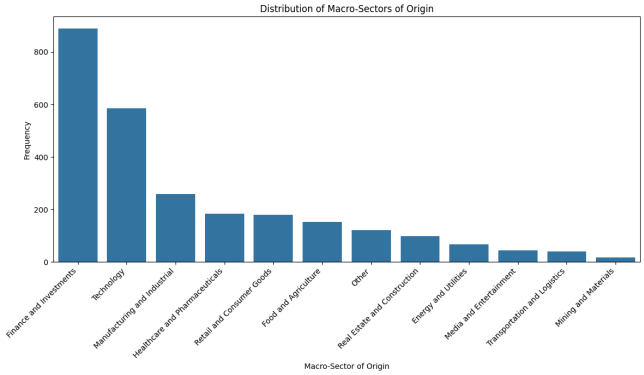


Figure 4.5. Bar plot of source showing the distribution of the new macro-sources.

5) Add Column Number Of Companies for each Billionaire

As part of the preprocessing phase, an enrichment step was performed to enhance the original dataset with additional information not initially available. Specifically, a new variable, number_of_companies, was introduced to estimate how many companies are associated with each individual. This feature provides a more complete view of each billionaire's business involvement and complements the existing attributes in the dataset.

To construct this variable, a Python script was developed to retrieve external information from Wikidata, a structured and publicly available knowledge base. The process uses the personName attribute to query the Wikidata API and identify the corresponding entity for each individual. Once the entity is found, relevant properties are extracted, focusing on relationships that link a person to organizations, such as founder, CEO, chairperson, manager, owner, or employee. Each identified organization is considered an associated company.

Several preprocessing steps were applied to ensure the reliability of the results. Names were cleaned by removing suffixes such as "& family" and trimming unnecessary characters to improve matching accuracy. Additionally, the script incorporates error handling mechanisms, including retries for failed requests, timeout management, and controlled delays between API calls. These measures reduce the impact of connection issues and help maintain a stable data collection process. Duplicate companies are removed before counting, ensuring that each organization is only considered once per individual.

After retrieving and processing the information, the number of associated companies for each individual was computed and added as a new column (number_of_companies). The enriched dataset was then exported to a new file, making it ready for subsequent analysis and data mining tasks.

An initial exploration of the number_of_companies variable was conducted to understand its distribution. The results indicate that most individuals are associated with a small number of companies, while a few cases present significantly higher values, suggesting the presence of outliers.

TABLE 4.4
NUMBER OF COMPANIES FREQUENCY

Number of Companies	Frequency
1	1933
2	538
3	53
4	1
5	1
6	1
7	1
10	105
15	1
20	1
60	1
75	1
100	1

The following figure illustrates the frequency distribution of the number of companies associated with each individual.

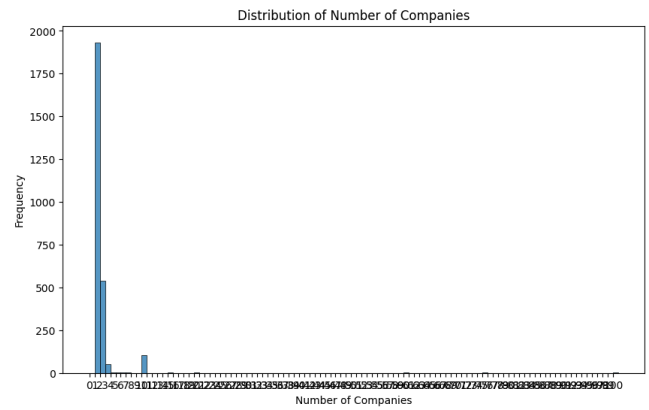


Figure 4.6. Distribution of Frequency of Distribution of add Columns Number of Companies.

B. Level 2 - Aggregation by Country

The data preparation process was carried out using Python and the Pandas library, applying grouping and aggregation operations on the *country* variable. Based on this transformation, three main types of features were constructed:

1) Wealth Structure

For each country, the following metrics were calculated:

- **total_billionaires**: total number of billionaires in the country.
- **selfmade_ratio**: proportion of billionaires who built their wealth themselves.
- **inherited_ratio**: proportion of inherited wealth, calculated as *1 – selfmade_ratio*.

These variables make it possible to analyze whether wealth in a country primarily comes from entrepreneurship and innovation or from inheritance.

2) Demographic Structure

The demographic characteristics of billionaires were analyzed through:

- **avg_age**: average age of billionaires in each country.
- **std_age**: standard deviation of age.
- **young_ratio**: proportion of billionaires under 50 years old.

These metrics help identify the degree of generational renewal and the economic dynamism of each country.

3) Sectoral Structure

The distribution of billionaires by industry and country was calculated using a cross-tabulation. This makes it possible to identify which economic sectors concentrate the highest wealth generation in each country, such as technology, finance, energy, or manufacturing.

4) Dominant Industry

For each country, the industry with the highest number of billionaires was identified. This was done by counting the frequency of billionaires by sector within each country and selecting the sector with the highest presence.

- **dominant_industry**: economic sector with the largest number of billionaires in each country.

This variable helps identify the main driver of wealth creation within each national economy.

5) Sectoral Concentration

To measure the degree of wealth concentration across different economic sectors, the Herfindahl–Hirschman Index (HHI) was used.

- **industry_concentration**: sectoral concentration index of wealth.

A high value indicates that wealth is concentrated in a few sectors, while a low value suggests a more diversified economic structure.

6) Feature Integration

Finally, all generated variables were integrated into a single country-level dataset using merge operations, with *country* as the common key.

The final dataset includes variables related to wealth structure, demographic structure, and sectoral structure.

B. Level 3.1 Dataset preparation for association task

1) Discretization of Variables

The first step in the discretization process consists of identifying the continuous variables in the dataset in order to evaluate which of them may benefit from transformation. Discretization converts numerical attributes into categorical intervals, which simplifies the analysis and enhances the interpretability of patterns discovered during the data mining process.

To support this step, variables with a continuous numerical type (float64) were explored through descriptive statistics, allowing an understanding of their distribution, central tendency, and variability. Based on this analysis, a subset of relevant variables was selected for further study: age, gdp_country, life_expectancy_country, gross_tertiary_education_enrollment, and population_country.

The statistical summary of these variables reveals important characteristics:

- gdp_country and population_country present very large ranges and high variability, indicating significant dispersion and possible skewness. These properties suggest that discretization methods such as quantile-based binning may be more appropriate to balance the distribution across intervals.
- age shows a moderate range with values concentrated around the mean, making it suitable for simpler discretization approaches such as equal-width binning.
- life_expectancy_country exhibits low variability and a relatively narrow range, which allows for the definition of meaningful intervals based on domain knowledge.
- gross_tertiary_education_enrollment presents moderate dispersion, making it a flexible candidate for either equal-width or equal-frequency discretization depending on the desired level of granularity.

This focused analysis provides a solid basis for selecting appropriate discretization techniques tailored to the characteristics of each variable, ensuring both interpretability and effectiveness in subsequent data mining tasks.

- Discretization of the variable Age

The age variable was discretized using a domain-based approach, in which continuous values are grouped into semantically meaningful age ranges. This method was selected because age naturally aligns with life-stage categories, making the resulting intervals easier to interpret and analyze.

Unlike purely statistical methods such as equal-width or equal-frequency binning, domain-based discretization produces categories that are more intuitive and meaningful from a real-world perspective. This is particularly important in association rule mining, where the interpretability of the discovered patterns is essential. By grouping individuals into well-defined age ranges, the generated rules can more clearly reveal relationships between age groups and other attributes in the dataset, such as industry, source of wealth, or self-made status.

Furthermore, using semantically grounded intervals helps avoid arbitrary partitions that may not reflect meaningful differences between individuals. As a result, this approach enhances both the interpretability and the practical relevance of the patterns identified during the data mining process.

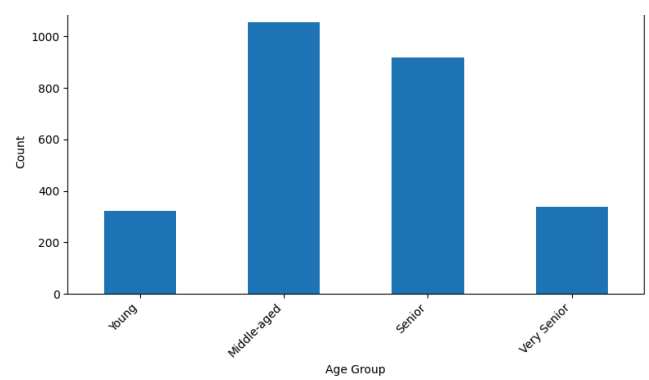


Figure 4.7. Frequency distribution after age variable discretization.

- Discretization of the variable GDP_Country

The gdp_country variable was discretized using quantile-based discretization (quartiles). This method divides the data into intervals that contain approximately the same number of observations, ensuring a balanced distribution across categories. This helps prevent situations where some intervals contain very few instances while others concentrate the majority of the data.

The resulting categories (Low GDP, Medium GDP, High GDP, and Very High GDP) provide an interpretable grouping of the data while maintaining a uniform distribution of observations. This balance improves the effectiveness of association rule mining by increasing the likelihood of discovering meaningful and statistically significant patterns.

The discretization was implemented using pd.qcut, which is particularly suitable for variables with skewed distributions. In contrast to equal-width binning, which can produce highly imbalanced intervals, this approach ensures that each bin contains a similar number of observations, leading to more reliable pattern extraction.

Additionally, the implementation includes the parameter duplicates='drop' to handle cases where some quantile boundaries are identical due to repeated values. In such situations, pd.qcut may generate fewer bins than initially specified. To maintain consistency, the number of labels is dynamically adjusted to match the actual number of bins created, ensuring a robust discretization process.

The figure below shows the new distribution of the gdp_country variable after discretization.

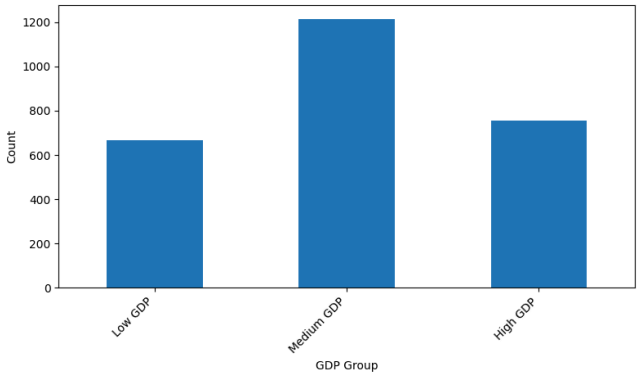


Figure 4.8. Frequency distribution after gdp_country variable discretization.

- Discretization of the variable Life Expectancy of the Country

The life_expectancy_country variable was discretized to enable its use in the association rule mining process, which requires categorical attributes. Transforming this continuous variable into intervals allows the identification of patterns based on grouped values rather than small numerical differences.

Discretization also improves interpretability by grouping values into broader categories, making the resulting patterns easier to analyze and understand. Additionally, it reduces the impact of minor variations between observations, allowing the analysis to focus on more general differences across groups.

For this variable, an equal-width binning approach was selected. This method divides the range of values into intervals of equal size, which is appropriate given the relatively narrow distribution of life expectancy compared to other variables in the dataset. This results in clear and well-separated categories without introducing significant imbalance.

The resulting intervals, Low (<75), Medium (75–78), High (78–81), and Very High (>81), provide an interpretable grouping while preserving meaningful differences between observations.

By converting the continuous values into these categories, the discretization process enhances the clarity and usefulness of the patterns discovered during association rule mining.

The figure below shows the new distribution of the life_expectancy_country variable after discretization.

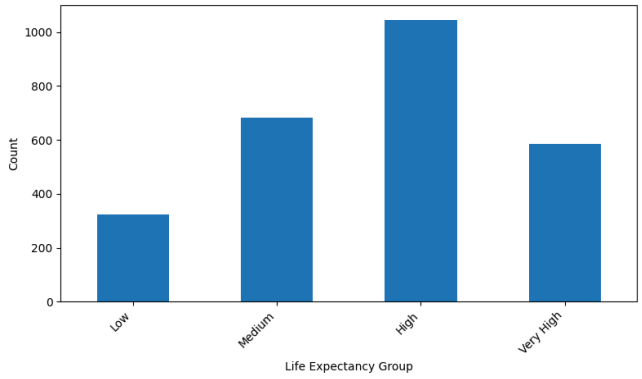


Figure 4.9. Frequency distribution after life_expectancy_country variable discretization.

- Discretization of the variable Gross Tertiary Education Enrollment

The gross_tertiary_education_enrollment variable was discretized to enable its use in the association rule mining process, which requires categorical attributes. Transforming this continuous variable into intervals allows the identification of patterns based on grouped values rather than small numerical differences.

Discretization improves interpretability by grouping observations into broader categories, making the resulting patterns easier to analyze. It also reduces the impact of minor variations in enrollment percentages, allowing the analysis to focus on more meaningful differences across groups.

For this variable, a quantile-based discretization approach (quartiles) was applied using pd.qcut. This method divides the data into intervals containing approximately the same number of observations, ensuring a balanced distribution across categories. This is particularly useful for avoiding imbalanced bins that could negatively affect the quality of the discovered patterns.

The resulting categories (Low education enrollment, Medium, High, and Very High) provide an interpretable grouping while maintaining a uniform distribution of observations across intervals.

Additionally, the implementation includes the parameter duplicates='drop' to handle cases where quantile boundaries are not unique due to repeated values. In such situations, the number of generated bins may be lower than expected, so the labels are dynamically adjusted to match the actual number of intervals, ensuring a consistent and robust discretization process.

The figure below shows the new distribution of the gross_tertiary_education_enrollment variable after discretization.

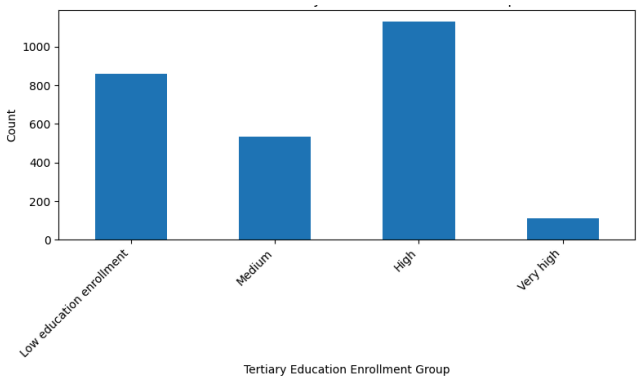


Figure 4.10. Frequency distribution after gross_tertiary_education_enrollment variable discretization.

- Discretization of the variable Population of the Country

The population_country variable was transformed into categorical intervals to make it suitable for association rule mining. Given its highly skewed nature (where a few countries concentrate extremely large populations) applying discretization directly on the original values would likely result in poorly balanced groups.

To mitigate this issue, a logarithmic transformation (base 10) was applied prior to discretization. This step reduces the effect of extreme values by compressing the upper range of the distribution, allowing for a more even spread of the data and facilitating the creation of meaningful intervals.

Once the data was transformed, quantile-based binning was performed using pd.qcut. This technique partitions the data into groups with a similar number of observations, which helps maintain balance across categories and supports the extraction of more reliable association rules.

The final grouping consists of four categories: Small population, Medium population, Large population, and Very large population. These categories offer a clear and interpretable segmentation of the data while preserving sufficient observations within each group.

Choosing four intervals provides an appropriate level of detail without overcomplicating the analysis. This ensures that relevant differences between groups are captured while maintaining enough support for meaningful pattern discovery.

By combining a logarithmic transformation with quantile-based discretization, the resulting representation of the variable is both stable and suitable for downstream analysis.

The figure below shows the new distribution of the population_country variable after discretization.

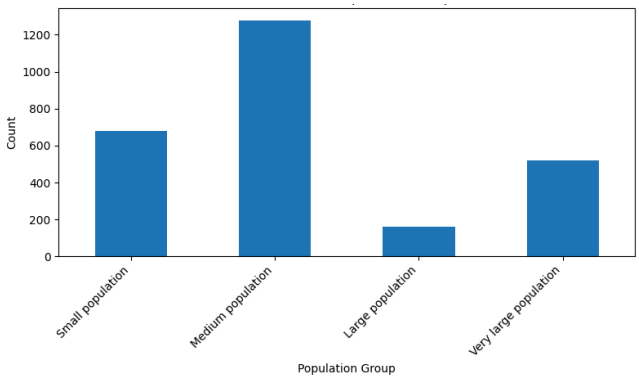


Figure 4.11. Frequency distribution after population_country variable discretization.

2) Binarization of Variables

Once the discretization stage is completed, it is necessary to transform the dataset into a representation compatible with association rule mining techniques. Unlike other analytical methods, these algorithms require the data to be structured in a transactional format, where each observation is described through the presence or absence of specific attributes.

To achieve this, a binarization process is applied as part of the preprocessing phase. This process consists of converting categorical and previously discretized variables into a set of binary attributes. Each possible value of a variable is represented as an independent feature, taking the value of 1 if the condition is satisfied and 0 otherwise. In this way, each record in the dataset is transformed into a collection of items that can be analyzed by the mining algorithm.

This transformation is fundamental for the correct functioning of algorithms such as Apriori, as it enables the identification of frequent itemsets and the extraction of association rules based on co-occurrence patterns. Moreover, the binary representation facilitates the interpretation of the results, since the discovered rules explicitly describe relationships between well-defined attribute-value pairs.

The variables selected for binarization are those that are categorical or have been previously discretized, as they can be naturally represented as binary features without losing interpretability.

The main candidates include: category, country, city, source, industries, countryOfCitizenship, status, gender, title, associated_companies, age_group, gdp_group, life_expectancy_group, tertiary_education_group, and population_group.

These variables represent meaningful attribute-value pairs that can be directly transformed into binary indicators, allowing each observation to be expressed as a set of items. This makes them suitable for association rule mining, as they facilitate the identification of frequent patterns and relationships within the data.

- Binarization of the variable Category

The category variable was selected for binarization due to its categorical nature and moderate cardinality, with 18 unique values across the dataset. This number of categories is appropriate, as it allows sufficient representation of each group without generating excessive sparsity.

A one-hot encoding approach was applied, where each category was transformed into an independent binary feature (e.g., category_Technology, category_Healthcare), taking the value of 1 when the category is present and 0 otherwise. As a result, 18 new binary columns were created, replacing the original variable.

This representation enables the model to treat each category as an individual item, facilitating the discovery of associations between different sectors and other attributes in the dataset.

- Binarization of the variables Country and Country of CitizenShip

Both country and countryOfCitizenship variables were evaluated for binarization; however, they exhibit high cardinality, with 79 and 77 unique values respectively. Transforming them directly using one-hot encoding would result in a large number of binary features, considerably increasing the dimensionality of the dataset.

This high number of categories leads to sparse representations, as many countries appear only a few times. Consequently, most of the generated binary features would have very low support, making them unlikely to contribute to frequent itemsets or meaningful association rules. Additionally, the increased dimensionality negatively impacts the computational efficiency of the mining algorithms.

Moreover, both variables provide highly specific information at the country level, which limits their ability to capture generalizable patterns across the dataset.

For these reasons, neither variable was binarized in its original form. A more appropriate approach would be to group countries into broader categories, such as geographic regions or economic classifications, in order to reduce cardinality and improve the quality of the analysis.

- Binarization of the variable City

The city variable was evaluated as a candidate for binarization; however, it exhibits very high cardinality, with 742 unique values. Most of these values appear only a few times, and in many cases are associated with a single observation.

Applying one-hot encoding would generate a large number of binary features with extremely low support, leading to a highly sparse dataset. This negatively impacts the performance of association rule mining algorithms, as such features are unlikely to contribute to frequent itemsets or meaningful rules.

Additionally, city-level information is highly specific and does not typically capture generalizable patterns relevant to the analysis.

For these reasons, the city variable was excluded from the binarization process.

- Binarization of the variable source

The source variable presents a limited number of distinct values (12 categories), making it a suitable candidate for binarization. Its relatively low cardinality ensures that each category maintains a reasonable number of observations, avoiding sparsity issues in the resulting dataset.

Another important aspect is that the categories are already grouped into broad and standardized sectors, which simplifies the transformation process and preserves meaningful distinctions between different sources of wealth.

To perform the transformation, one-hot encoding was applied, generating 12 new binary features (e.g., source_Technology, source_Energy and Utilities), where each feature indicates the presence or absence of a specific category. The original variable was subsequently replaced by these binary columns.

This representation allows the model to incorporate the source of wealth as a set of independent items, supporting the identification of associations with other attributes in the dataset.

- Binarization of the variable industries

The industries variable presents a low number of distinct categories (around 10), which makes it suitable for binarization. This limited cardinality ensures that the resulting binary features remain manageable and avoids excessive sparsity in the dataset.

Additionally, the categories correspond to broad industry sectors, which guarantees sufficient representation across observations. This increases the likelihood that the generated binary attributes will meet the minimum support required for association rule mining.

The transformation was carried out using one-hot encoding, creating a set of binary features (e.g., industries_Technology & Telecommunications, industries_Healthcare). In total, 10 new columns were generated, replacing the original variable.

This encoding allows each industry to be treated as an independent item, facilitating the discovery of relationships between sectors and other attributes in the dataset.

- Binarization of the variable Status

The status variable contains only 6 distinct categories, making it a suitable candidate for binarization. Its low cardinality ensures that the transformation results in a small number of additional features, without significantly increasing the dimensionality of the dataset.

One-hot encoding was applied, generating 6 binary columns corresponding to each category: status_U, status_D, status_N, status_Split Family Fortune, status_E, and status_R. Each column indicates the presence (1) or absence (0) of the corresponding status for a given observation.

Due to the limited number of resulting features, this transformation remains efficient and does not introduce sparsity issues. It also allows each category to be treated as an independent item in the association rule mining process.

- Binarization of the variable Gender

The gender variable contains only two possible values: M and F, which makes it a binary attribute by nature. This very low cardinality allows for a straightforward transformation without increasing the complexity of the dataset.

One-hot encoding was applied, resulting in two binary columns: gender_M and gender_F. Each column indicates whether an individual belongs to the corresponding category.

Given that only two features are generated, this transformation does not introduce sparsity or dimensionality issues. It also ensures that the variable can be directly incorporated into the association rule mining process as a clear and interpretable attribute.

- Binarization of the variable Title

The title variable initially appears to have very high cardinality, with 210 unique values. However, this is due to the fact that each entry represents a combination of multiple roles (e.g., Founder & CEO, Chairman & Owner), rather than a single category.

After preprocessing the variable (including text normalization and splitting combined roles) a total of 149 unique terms were identified. This confirms that the variable is composed of recurring roles such as founder, entrepreneur, CEO, and chairman, which appear with high frequency (e.g., founder appears more than 1700 times, and entrepreneur over 1500 times).

Despite this decomposition, the number of resulting terms remains high, and the variable includes a mix of relevant roles and noisy or context-specific tokens (e.g., facebook, oil, pipes, inc.). This leads to a heterogeneous representation with many low-frequency elements.

As a result, binarizing this variable would generate a large number of sparse features, many of them with very low support, which would negatively impact the efficiency of the mining process and the quality of the extracted rules.

For these reasons, the title variable was not binarized in its current form and would require additional filtering or abstraction to be effectively used.

- Binarization of the variable Associated_companies

The associated_companies variable exhibits extremely high cardinality, with 1,099 unique values. This indicates that most entries correspond to specific or rarely repeated company names.

Applying one-hot encoding in this case would produce a very large number of binary features, most of them with very low frequency. Such sparse features are unlikely to meet the minimum support thresholds required for association rule mining, limiting their contribution to meaningful pattern discovery.

In addition, this transformation would significantly increase the dimensionality of the dataset, negatively impacting computational efficiency and making the results harder to interpret.

It is also important to note that company names tend to behave as identifiers rather than generalizable attributes, which further reduces their usefulness in this type of analysis.

For these reasons, the associated_companies variable was excluded from the binarization process.

- Binarization of the Discretized Variables: Age, GDP_Country, Life_Expectancy, Tertiary_Education, Population

age_group: The age_group variable contains 4 categories (Young, Middle-aged, Senior, Very Senior), making it a low-cardinality attribute. One-hot encoding was applied, generating 4 binary columns (age_group_Young, age_group_Middle-aged, age_group_Senior, age_group_Very Senior).This transformation preserves interpretability while allowing each age segment to be analyzed independently.

gdp_group: The gdp_group variable includes 3 categories (Low GDP, Medium GDP, High GDP), resulting in a very compact representation. After binarization, 3 binary columns were created. Due to the small number of groups, this transformation does not introduce sparsity and remains efficient for pattern discovery.

life_expectancy_group: The life_expectancy_group variable is composed of 4 ordered categories (Low, Medium, High, Very High). One-hot encoding produced 4 binary features. This allows each level to be treated as a separate condition, facilitating the analysis of relationships with other variables.

tertiary_education_group: The tertiary_education_group variable contains 4 categories (Low education enrollment, Medium, High, Very high). After binarization, 4 binary columns were generated. The limited number of categories ensures good representation across groups and avoids sparsity issues.

population_group: The population_group variable also consists of 4 categories (Small, Medium, Large, Very large population). One-hot encoding resulted in 4 binary features. This representation simplifies a highly skewed variable while maintaining meaningful distinctions between groups.

Initially, the dataset contained 27 attributes, combining numerical, categorical, and identifier-type variables. After the preprocessing stage, which included discretization and subsequent binarization, the structure of the dataset changed significantly. Through one-hot encoding, each selected categorical or discretized variable was expanded into multiple binary columns, increasing the dimensionality of the data. As a result, the final dataset contained more than 80 attributes, each representing a specific category or condition in binary form.

In total, 9 variables were effectively binarized: category, source, industries, status, gender, age_group, gdp_group, life_expectancy_group, tertiary_education_group, and population_group. These variables were chosen because they exhibit low cardinality, clear semantic meaning, and sufficient representation across observations, making them suitable for association rule mining. In contrast, variables such as country and countryOfCitizenship (over 70 unique values each), city (742 unique values), associated_companies (1099 unique values), and title (more than 200 combinations and 149 decomposed terms) were excluded due to their high cardinality and the sparsity they would introduce if binarized. Additionally, attributes like personName, birthDate, and date were not considered, as they function as identifiers or temporal variables rather than meaningful inputs for pattern discovery.

C. Level 3.2 Dataset preparation for classification task

To perform the classification of countries into Innovation Ecosystem, Rentier Paradise, or Extraction Economy, it was necessary to transform the raw billionaire dataset into a country-level structural dataset. Since the original data describes individuals, the first step consisted of aggregating the information by country and generating features that capture the structural characteristics of each national wealth ecosystem.

This transformation process aimed to produce variables that reflect three key dimensions described in the project objectives:

1. Sectoral structure of wealth generation
2. Sources of wealth
3. Demographic characteristics of the billionaire population

These dimensions are conceptually aligned with the idea that the structure of elite wealth may signal broader national economic patterns.

The first step consisted of grouping the dataset by **country**, since the classification model operates at the **national level** rather than the individual level. We use the dataset built on **level 2.**

Also, to characterize the **sectoral Composition of wealth**, it was computed the industry shares; i. e for each country, the share of billionaires associated with each industry was calculated:

$$Share_{industry} = \frac{billionaires\ in\ industry}{total\ billionaires\ in\ country}$$

These variables capture the sectoral distribution of wealth generation.

This is particularly important for distinguishing the conceptual categories of the project:

- Innovation Ecosystem: Higher share of technology or knowledge-driven sectors.
- Extraction Economy: High concentration in natural resource industries.
- Rentier Paradise: High presence of finance, real estate, or rent-seeking sectors.

In addition to industry sectors, the dataset includes the source of wealth (source). the following features were derived:

- Source Shares: The proportion of billionaires associated with each source category was calculated relative to the total number of billionaires in each country. These features help capture how wealth is generated structurally, distinguishing between different mechanisms such as entrepreneurship, inheritance, or sector-specific wealth accumulation.

- Dominant Source: the dominant source variable identifies the most frequent origin of wealth in each country, providing a simplified indicator of the primary wealth generation mechanism.

- Source Concentration: Similar to the industry analysis, a concentration index for wealth sources was computed using the squared shares of each source category.

This variable captures whether wealth creation is diversified across multiple sources, or dominated by a single mechanism.

3. Normalization of Numerical Variables

After completing data cleaning and imputation, a normalization step was applied to ensure that all numerical variables are on a comparable scale. This is particularly important for downstream analytical tasks such as clustering, distance-based methods, and machine learning models, where differences in magnitude between variables can bias the results.

The dataset contains multiple numerical attributes with significantly different ranges. For instance, variables such as gdp_country and population_country have very large magnitudes, while others such as cpi_change_country or tax_revenue_country_country are expressed as percentages with relatively small values. Without normalization, variables with larger scales would dominate the analysis.

To address this issue, Min-Max normalization was applied to a selected set of numerical variables. This technique rescales each variable to a fixed range of [0,1][0, 1][0,1], preserving the relative relationships between values while eliminating scale differences.

The following variables were normalized:

- *finalWorth*
- *age*
- *cpi_country*
- *cpi_change_country*
- *gdp_country*
- *gross_tertiary_education_enrollment*
- *gross_primary_education_enrollment_country*
- *life_expectancy_country*
- *tax_revenue_country_country*
- *total_tax_rate_country*
- *population_country*
- *number_of_companies*

V. Association task

This section presents the implementation of association rule mining techniques applied to the dataset in order to identify structural relationships between wealth accumulation and national well-being indicators. Given the nature of the association task, the original dataset required several preprocessing and transformation steps to adapt it into a transactional format suitable for association analysis. In particular, continuous variables related to economic, social, and industrial indicators were discretized into categorical ranges, allowing each country to be represented as a set of items describing both its dominant wealth-generation sectors and its socioeconomic conditions.

To perform the association analysis, two well-known algorithms were implemented: Apriori and FP-Growth. Both algorithms were used to discover frequent itemsets and generate association rules capable of revealing recurring patterns between sectors of wealth generation and indicators such as life expectancy, fiscal revenues, inflation behavior, educational enrollment, and other measures of national well-being. The use of both approaches also enabled a comparison in terms of efficiency and rule generation capabilities, especially considering the dimensionality and sparsity of the transformed transactional dataset.

Through this implementation, the project seeks to uncover hidden structural associations that may reflect broader socioeconomic dynamics and potential trade-offs between industrial specialization and human development indicators.

A. *Preprocessed Dataset Transformation into a Binary Transactional Representation*

The preprocessing pipeline was designed to transform the original raw dataset into a binary transactional representation suitable for association rule mining using the Apriori and FP-Growth algorithms. Since association rule mining operates on transactions composed of discrete items rather than continuous numerical variables, several feature engineering, aggregation, and discretization procedures were required to adapt the dataset to the objectives of the association task.

The first transformation focused on modeling the sectoral and source composition of wealth generation within each country. For this purpose, the proportion of each industry sector and wealth source was calculated relative to the total billionaire wealth structure of a country. These proportions were then discretized using quantile-based binning into categorical levels such as low, mid, and high. Countries without representation in a specific industry or source were initially assigned a none category. This process allowed the dataset to capture not only the existence of sectors, but also their relative dominance within national wealth accumulation patterns.

Additional variables were created to characterize the internal structure of wealth at the country level. Metrics such as average billionaire wealth, number of billionaires, wealth per capita, and the ratio between billionaire wealth and GDP were computed and later categorized into discrete ranges. Some variables employed standard quantile-based discretization, while others required manually defined thresholds in order to preserve economically meaningful distinctions between countries.

To represent wealth concentration and inequality, the proportion of total billionaire wealth controlled by the richest individual in each country was calculated. This indicator was categorized into levels ranging from low concentration to oligarchic concentration using economically motivated thresholds. In parallel, economic diversification was measured through Shannon entropy applied to the sector distribution of each country, enabling economies to be classified according to their degree of diversification.

The dataset was further enriched with variables describing the origin of wealth generation. In particular, the proportion of self-made billionaires within each country was computed and discretized into categorical levels, allowing the analysis to distinguish between countries dominated by inherited wealth structures and those characterized by entrepreneurial wealth creation.

Macroeconomic and social indicators were also incorporated into the transactional representation. Variables such as life expectancy, CPI variation, GDP, tax-related indicators, population, and educational metrics were aggregated at the country level and transformed into categorical intervals. Different preprocessing strategies were applied depending on the statistical behavior and economic interpretation of each variable. For example, GDP and population variables were log-transformed prior to discretization, while CPI-related indicators used fixed thresholds designed to better reflect inflationary conditions.

After all variables had been transformed into categorical attributes, they were consolidated into a single dataset. Missing values were replaced using categories such as unknown or 0 to guarantee dataset completeness and avoid inconsistencies during the association mining process. Finally, One-Hot Encoding was applied to all categorical variables, converting the dataset into a binary transactional matrix where each column represented the presence or absence of a specific characteristic. Categories associated with none or unknown values were subsequently removed in order to reduce noise and improve the interpretability of the generated association rules.

B. Apriori Algorithm

The Apriori algorithm is one of the most widely used techniques for discovering frequent itemsets and association rules within transactional datasets. Its main objective is to identify combinations of attributes that frequently occur together and subsequently derive meaningful association rules from these patterns. Unlike FP-Growth, Apriori follows an iterative candidate-generation strategy in which larger itemsets are progressively constructed from smaller frequent itemsets while pruning combinations that do not satisfy the minimum support threshold. Although this approach can be computationally expensive for large datasets, it provides a clear and interpretable framework for association rule mining.

For this implementation, a transactional dataset at the country level was constructed by transforming multiple socioeconomic, wealth, and industrial indicators into categorical variables through quantile-based binning and domain-specific thresholds. The resulting features included variables associated with wealth concentration, number of billionaires, sector diversification, self-made wealth ratios, industrial participation, tax structures, education indicators, life expectancy, and CPI variation. After preprocessing, the dataset was converted into a one-hot encoded binary representation, where each column indicated the presence or absence of a specific categorical attribute. This transformation made the dataset directly compatible with the Apriori algorithm implemented through the `apriori` function from the `mlxtend.frequent_patterns` library.

Frequent itemsets were generated using a minimum support threshold of 0.1, meaning that only attribute combinations appearing in at least 10% of the countries were retained as relevant patterns. Subsequently, association rules were extracted using the `association_rules` function with `metric="lift"` and `min_threshold=1`, ensuring that only positively correlated associations were considered for analysis. This configuration enabled the identification of statistically meaningful relationships between structural wealth indicators and macroeconomic country characteristics.

To improve interpretability and maintain analytical consistency, several filtering stages were applied to the generated rules. First, the antecedents were restricted to variables directly related to billionaire wealth structures, including industrial shares, wealth concentration measures, wealth-per-capita indicators, sector entropy, and self-made wealth ratios. Consequents, on the other hand, focused primarily on macroeconomic and well-being indicators such as life expectancy, CPI variation, tertiary education enrollment, tax revenue, and total tax rate. Additional constraints limited antecedents to a maximum of two items and consequents to exactly one item, preventing overly complex rules and facilitating interpretation. Furthermore, all rules containing the intermediate category `mid` were removed from both antecedents and consequents in order to emphasize stronger and more distinctive structural patterns.

Through this process, the Apriori algorithm generated a refined set of association rules that revealed recurring relationships between wealth-generation structures and national socioeconomic conditions. These rules provided insights into how factors such as economic diversification, concentration of billionaire wealth, and industrial specialization may be associated with broader indicators of social and economic well-being across countries.

1) Algorithm Results

The total amount of rules found by FP-Growth algorithm were 47, to be continued we show some of the results obtained:

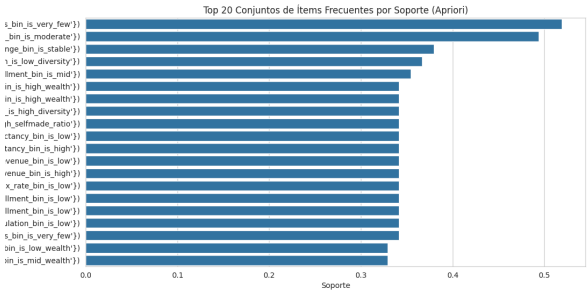


Figure 5.1. Top-20 Frequent Itemsets found by Aprior algorithm.

In Figure 5.1, we can observe the top 20 most frequent itemsets generated by the Apriori algorithm, ranked according to their support values. The horizontal bar chart illustrates the combinations of categorical attributes that appear most frequently across the country-level transactional dataset. The X-axis represents the support metric, which measures the proportion of countries in which a given itemset occurs, while the Y-axis lists the corresponding itemsets identified during the frequent pattern mining process. The results reveal that the most common patterns are strongly associated with categories such as `num_billionaires_bin_is_very_few`, `cpi_change_bin_is_moderate`, and `sector_entropy_bin_is_low_diversity`, indicating that many countries in the dataset tend to have relatively few billionaires, moderate inflation changes, and limited economic diversification. Additionally, variables related to wealth structure and education, including `average_wealth_bin_is_high_wealth`, `wealth_per_capita_bin_is_high_wealth`, and `gross_tertiary_education_enrollment_bin_is_mid`, also appear with high support, suggesting that these socioeconomic characteristics frequently coexist within the

analyzed countries. The figure further highlights the simultaneous presence of both low and high categories for indicators such as life expectancy and tax revenue, reflecting the heterogeneous economic and social conditions represented in the dataset. Overall, these frequent itemsets provide an initial overview of the dominant structural and macroeconomic patterns before the extraction and interpretation of more specific association rules.

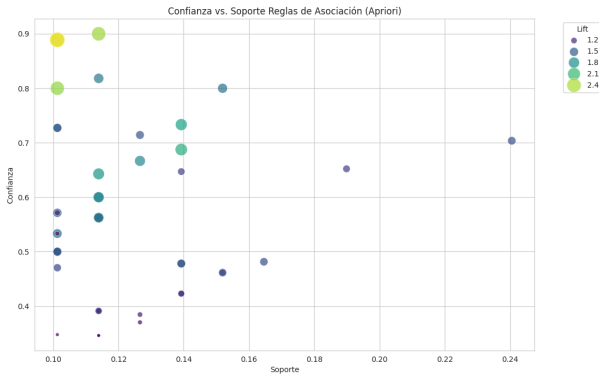


Figure 5.2. Confidence vs Support of Association Rules found by Apriori algorithm.

In Figure 5.2, we can observe the relationship between support and confidence for the refined association rules generated using the Apriori algorithm. The scatter plot represents each association rule as a bubble, where the X-axis corresponds to the support value and the Y-axis represents the confidence of the rule. Additionally, the size and color intensity of each bubble indicate the lift metric, allowing a simultaneous visualization of the statistical strength and relevance of the discovered associations. Rules with larger and brighter bubbles correspond to higher lift values, indicating stronger positive relationships between antecedents and consequents.

The distribution of the rules reveals that most associations are concentrated within support values between 0.10 and 0.15, suggesting that many meaningful patterns appear in approximately 10% to 15% of the analyzed countries. Despite their moderate support, several rules exhibit high confidence values ranging from 0.70 to 0.90, indicating that when the antecedent conditions are satisfied, the consequent socioeconomic conditions are highly likely to occur. This behavior reflects the existence of relatively consistent structural relationships between wealth-generation characteristics and macroeconomic indicators.

Furthermore, some rules achieve both high confidence and relatively high lift values simultaneously, highlighting particularly strong associations that are unlikely to occur by random chance. These rules may correspond to specific combinations of wealth concentration, industrial specialization, or self-made wealth structures that systematically coexist with indicators such as education enrollment, life expectancy, tax structures, or CPI stability. Conversely, rules with lower lift and confidence values suggest weaker or more generalized relationships that may be less structurally significant.

The figure also demonstrates the trade-off commonly observed in association rule mining between support and confidence. Rules with very high support tend to exhibit moderate confidence, while some highly confident rules appear with lower support, meaning they describe more specific but still statistically meaningful country profiles. Overall, the visualization provides a comprehensive overview of the quality and distribution of the refined Apriori association rules, facilitating the identification of the most relevant structural patterns within the dataset.

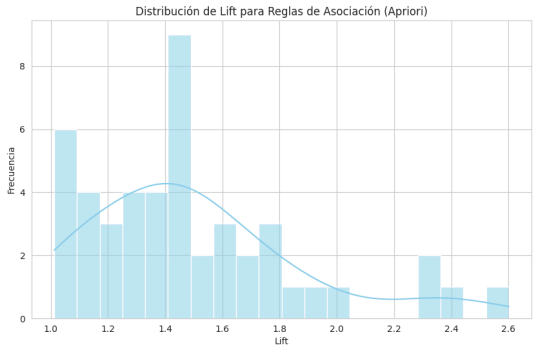


Figure 5.3. Lift Distribution for Association Rules found by Apriori algorithm.

In Figure 5.3,, we can observe the distribution of lift values for the refined association rules generated using the Apriori algorithm. The histogram illustrates the frequency of rules according to their lift metric, while the overlaid density curve provides a smoothed representation of the overall distribution pattern. The X-axis represents the lift values of the association rules, whereas the Y-axis indicates the frequency of occurrence for rules within each lift interval.

The distribution shows that most association rules are concentrated within lift values ranging approximately from 1.1 to 1.8, with a noticeable peak around 1.4 to 1.5. Since lift values greater than 1 indicate positive statistical dependence between antecedents and consequents, this result suggests that the majority of the extracted rules represent meaningful associations rather than random co-occurrences. In particular, lift values near 1.5 imply that the consequent is about 50% more likely to occur when the antecedent conditions are present compared to its baseline probability in the dataset.

Additionally, the histogram presents a right-skewed distribution, where a smaller number of rules achieve lift values above 2.0. These high-lift rules correspond to especially strong and distinctive structural relationships between wealth-related characteristics and socioeconomic indicators. Such associations may reflect highly specific country profiles in which factors like economic diversification, wealth concentration, or industrial specialization strongly correlate with macroeconomic outcomes such as education levels, inflation stability, or life expectancy.

The relatively limited number of rules with lift values close to 1 also indicates that the filtering process successfully removed many weak or non-informative associations. Overall, the figure demonstrates that the refined Apriori rules exhibit predominantly positive and statistically relevant relationships, supporting the effectiveness of the preprocessing and filtering strategy used during the association rule mining process.

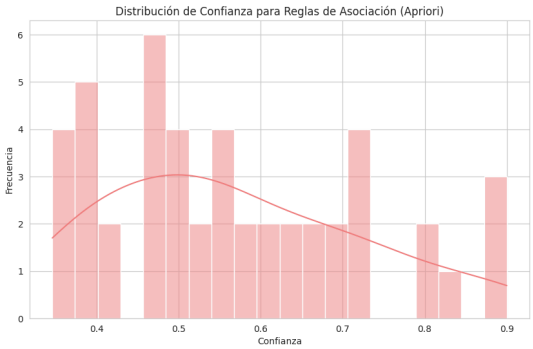


Figure 5.4. Confidence Distribution for Association Rules found by Aprior algorithm.

In Figure 5.4, we can observe the distribution of confidence values for the refined association rules generated using the Apriori algorithm. The histogram displays the frequency of rules according to their confidence levels, while the smooth density curve illustrates the general tendency of the distribution. The X-axis represents the confidence metric of the association rules, whereas the Y-axis indicates the number of rules falling within each confidence interval.

The distribution reveals that most rules are concentrated within confidence values ranging approximately from 0.45 to 0.70, with a noticeable peak around 0.50. This indicates that many of the extracted association rules possess moderate to high predictive reliability, meaning that when the antecedent conditions are satisfied, the consequent socioeconomic conditions occur with relatively high probability. In the context of association rule mining, these confidence levels suggest that the identified structural relationships between wealth-related attributes and macroeconomic indicators are not random, but instead represent recurring patterns within the dataset.

Additionally, the figure shows the presence of several rules with confidence values above 0.80 and approaching 0.90. These highly confident rules correspond to particularly strong associations in which the antecedent almost always implies the consequent. Such patterns may indicate highly consistent relationships between specific economic structures—such as wealth concentration, industrial specialization, or self-made wealth composition—and national well-being indicators including education, tax structures, CPI stability, or life expectancy.

The distribution also exhibits a slight right-skewed behavior, where fewer rules achieve extremely high confidence values. This is expected in association rule mining, as highly reliable rules are generally more difficult to obtain after applying strict filtering criteria. Overall, the figure demonstrates that the refined Apriori rules maintain relatively strong predictive consistency, supporting the validity of the extracted associations and reinforcing the interpretability of the resulting structural patterns within the dataset.

2) Relevant Association Rules Found

- Rule 1: Low manufacturing participation associated with CPI stability

Confidence: 0.90
Lift: 2.37

This rule indicates that 90% of the countries with low billionaire participation in manufacturing and industrial wealth sources exhibit stable CPI variation levels. The high lift value confirms that this relationship appears significantly more often than would be expected by chance. The pattern suggests that economies less dependent on industrial manufacturing may experience greater macroeconomic stability, potentially due to stronger diversification toward services, finance, or technology sectors that are less vulnerable to industrial supply-chain shocks and production volatility.

- Rule 2: Low industrial concentration reinforcing inflation stability

Confidence: 0.89
Lift: 2.34

This rule strengthens the previous finding by combining low industrial participation with low manufacturing-related wealth concentration. The association reveals that countries exhibiting both characteristics are highly likely to maintain stable inflation conditions. The elevated lift value indicates a strong structural relationship between reduced industrial dependence and controlled CPI variation. This may reflect economies where wealth generation is concentrated in less production-intensive sectors, reducing exposure to fluctuations in industrial costs and commodity prices.

- Rule 3: High technology and diversified “other” sectors associated with low tertiary education enrollment

Confidence: 0.89
Lift: 2.60

This was one of the strongest rules identified in the analysis. A confidence of 89% indicates that the relationship is highly reliable across the dataset, while the lift value suggests a very strong positive association. The rule implies that countries with high billionaire concentration in technology and “other” diversified sectors frequently exhibit low tertiary education enrollment. This may reveal economies where wealth generation is concentrated among specific entrepreneurial or technological elites without necessarily translating into widespread higher educational development or inclusive human capital growth.

- Rule 4: High participation of “other” sectors associated with low tertiary education enrollment

Confidence: 0.69
Lift: 2.01

This rule shows that countries with a strong billionaire presence in “other” economic sectors tend to exhibit lower tertiary education enrollment rates. Although the confidence is lower than in the previous rule, the lift value still indicates a statistically meaningful association. The pattern may suggest that some wealth-generation structures rely more heavily on non-traditional industries, resource-based activities, or concentrated economic niches where broad access to higher education is not the primary driver of wealth accumulation.

- Rule 5: Consumer discretionary wealth associated with high life expectancy

Confidence: 0.64
Lift: 1.88

This rule indicates that countries with high billionaire participation in consumer discretionary sectors are likely to exhibit high life expectancy levels. Consumer discretionary industries,

such as luxury goods, fashion, and premium retail, are generally concentrated in economically developed nations with mature consumer markets and advanced healthcare systems. The positive lift value suggests that these structural economic characteristics coexist with favorable social conditions and higher overall standards of living.

- Rule 6: Low financial-sector participation associated with high total tax rates

Confidence: 0.56
Lift: 1.71

This rule reveals that countries with low billionaire concentration in the financial and investment sectors are frequently associated with high total tax rates. Although the confidence level is moderate, the lift value confirms a meaningful positive association. The relationship may reflect economies characterized by stronger fiscal intervention, redistributive policies, or regulatory environments that limit the dominance of extreme financial wealth concentration while supporting broader public-sector financing.

- Rule 7: High financial and investment wealth associated with moderate inflation

Confidence: 0.82
Lift: 1.66

This rule indicates that countries where billionaire wealth is strongly concentrated in finance and investments tend to exhibit moderate CPI variation levels. The high confidence demonstrates that the relationship is relatively consistent throughout the dataset. The rule may reflect economies with active financial markets and continuous investment activity, where economic growth and capital circulation generate controlled but persistent inflationary pressures instead of extreme inflation instability.

- Rule 8: High self-made financial wealth associated with moderate inflation

Confidence: 0.80
Lift: 1.62

This rule suggests that countries with both a high proportion of self-made billionaire wealth and strong financial-sector participation are likely to experience moderate inflation levels. The relatively high confidence and positive lift value indicate a stable structural relationship. This pattern may imply that entrepreneurial and investment-driven economies generate sustained economic dynamism and market activity, contributing to moderate inflationary environments that reflect continuous expansion and financial development.

C. FP-Growth Algorithm

The FP-Growth (Frequent Pattern Growth) algorithm is an efficient method for discovering frequent itemsets in transactional datasets, which constitutes a fundamental task in association rule mining. Unlike the Apriori algorithm, which explicitly generates and evaluates candidate itemsets, FP-Growth relies on a compressed data structure known as the FP-Tree to store transactional information. This approach avoids the costly generation of candidate combinations and significantly improves computational efficiency, particularly when working with dense datasets or low minimum support thresholds.

For this implementation, the same transactional dataset used in the Apriori analysis, was employed. This dataset had previously been transformed into a binary format where each column represented the presence or absence of a specific categorical attribute associated with wealth structure or national socioeconomic indicators. As a result, the dataset was directly compatible with association rule mining techniques.

Frequent itemsets were identified using the fpgrowth function from the mlxtend.frequent_patterns library with a minimum support threshold of 0.1. Consequently, only itemsets appearing in at least 10% of the countries were considered frequent and retained for further analysis. Once the frequent itemsets were obtained, association rules were generated using the association_rules function from the same library. The rule generation process was configured with metric="lift" and min_threshold=1, ensuring that only rules exhibiting positive statistical association were considered.

To maintain consistency with the Apriori-based analysis and ensure interpretability of the results, several filtering criteria were applied to the generated rules. First, antecedents were restricted exclusively to variables describing the structure and origin of wealth accumulation, particularly those associated with sector participation and self-made wealth indicators. In contrast, consequents were limited to country-level well-being and macroeconomic indicators, including variables related to life expectancy, CPI variation, education, and fiscal conditions.

Additionally, the complexity of the rules was constrained by limiting antecedents to a maximum of two items and consequents to exactly one item. This restriction facilitated the interpretation of the extracted patterns while avoiding excessively complex associations. Finally, all rules containing the intermediate category mid in either antecedents or consequents were removed. The exclusion of intermediate categories aimed to emphasize clearer and more distinctive structural relationships between wealth-generation sectors and national socioeconomic conditions.

Through this process, FP-Growth produced a refined set of association rules that were subsequently analyzed and visualized to identify recurring structural patterns linking economic wealth structures with indicators of national well-being.

1) Algorithm Results

The total amount of rules found by FP-Growth algorithm were 47, to be continued we show some of the results obtained:

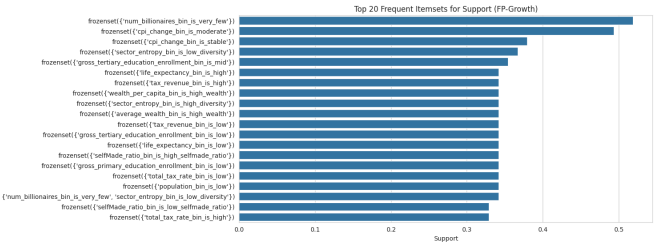


Figure 5.5. Top-20 Frequent Itemsets found by FP-Growth algorithm.

In Figure 5.5, we can observe the top 20 most frequent itemsets ranked by support. This bar chart presents the combinations of attributes that appear most frequently within the transactional dataset. The X-axis represents the support value, defined as the proportion of countries in which a given itemset appears, while the Y-axis displays the corresponding itemsets.

The results show that the most frequent itemsets commonly include characteristics such as num_billionaires_bin_is_very_few, wealth_per_capita_bin_is_low_wealth, and sector_entropy_bin_is_low_diversity. This indicates that a large proportion of countries in the dataset tend to exhibit a relatively

small number of billionaires, low wealth per capita, and limited economic diversification. In addition, categories such as cpi_change_bin_is_moderate and life_expectancy_bin_is_high also appear frequently, suggesting that moderate inflation levels and high life expectancy may coexist even in countries with lower concentrations of billionaire wealth.

The figure also reveals the frequent presence of several _is_low and _is_high categories associated with specific industries and wealth sources. This pattern suggests that many countries exhibit strong specialization or concentration in particular economic sectors, reinforcing the idea that wealth generation structures are often unevenly distributed across industries. Overall, the frequent itemsets provide an initial overview of the dominant structural characteristics represented in the dataset before the generation of association rules.

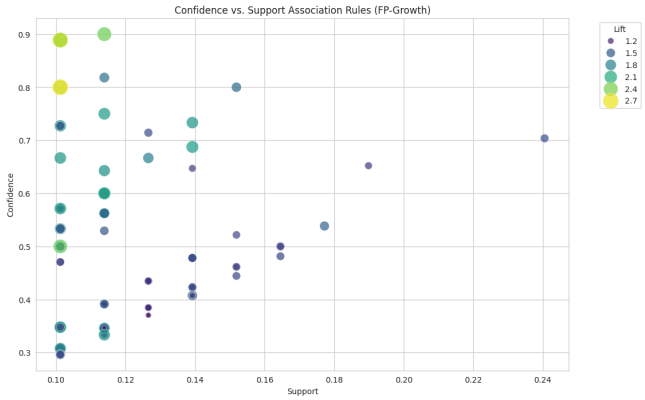


Figure 5.6. Confidence vs Support of Association Rules found by FP-Growth algorithm.

In Figure 5.6, we can observe a scatter plot representing the refined association rules generated during the analysis. Each point in the graph corresponds to a single association rule. The X-axis represents the support of the rule, which measures how frequently the antecedent and consequent occur together in the dataset, while the Y-axis represents the confidence of the rule, indicating the probability that the consequent occurs given the antecedent. Additionally, both the size and color intensity of each point encode the lift value of the rule, with larger and brighter points corresponding to stronger positive associations.

Most of the rules are concentrated in the lower-left region of the graph, indicating relatively low support and confidence values. This behavior is expected in association rule mining, particularly when analyzing complex socioeconomic relationships, since many combinations of structural wealth indicators and national well-being variables occur only in a limited subset of countries. Nevertheless, a noticeable group of rules appears distributed toward the upper and right sections of the figure, exhibiting higher support and confidence values along with elevated lift scores.

These rules are especially relevant because they combine statistical reliability with stronger associative significance. In particular, the rules with the highest lift values are generally located within the low-to-medium support range, suggesting that the most meaningful associations are not necessarily the most frequent ones. Instead, these rules capture less obvious but more informative structural relationships between patterns of wealth accumulation and socioeconomic indicators. Consequently, the figure highlights the importance of considering lift alongside support and confidence when identifying association rules with potentially valuable interpretative insights.

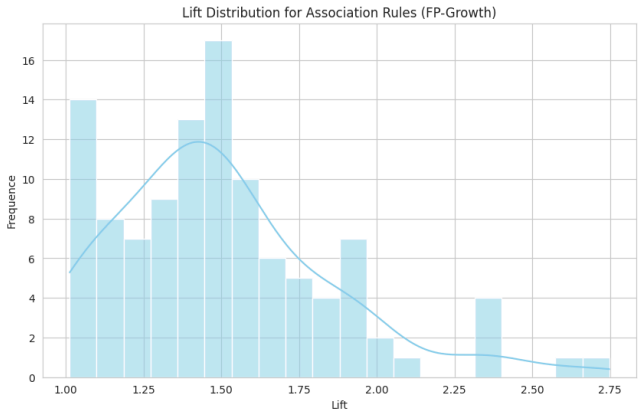


Figure 5.7. Lift Distribution for Association Rules found by FP-Growth algorithm.

In Figure 5.7, we can observe the distribution of lift values for all refined association rules generated during the FP-Growth analysis. The histogram displays lift values along the X-axis and the frequency of rules within each lift interval along the Y-axis. This visualization provides an overview of the overall strength of the associations discovered in the dataset.

The distribution is noticeably right-skewed, with the majority of the rules presenting lift values between 1 and approximately 1.5–2.0. This result is significant because a lift greater than 1 indicates a positive association between the antecedent and the consequent, meaning that both occur together more frequently than would be expected by random chance. The concentration of rules within this interval suggests that the algorithm successfully identified numerous meaningful relationships between wealth structure variables and national well-being indicators.

At the same time, the histogram exhibits a long tail extending toward higher lift values, reaching approximately 2.6. These higher lift values correspond to particularly strong and non-trivial association rules, indicating relationships with a substantially stronger dependency between antecedents and consequents. Although such rules are less frequent, they are often the most informative because they reveal structural patterns that are not immediately obvious from the dataset distribution alone.

Overall, the figure confirms the presence of statistically meaningful associations beyond simple coincidence and highlights the existence of a subset of highly significant rules that deserve further interpretation and analysis within the context of wealth accumulation and socioeconomic conditions.

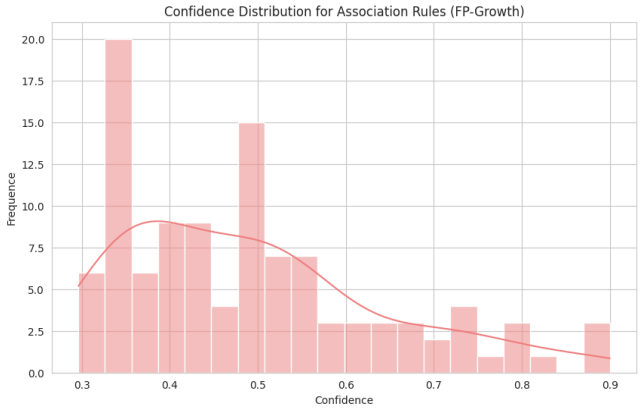


Figure 5.8. Confidence Distribution for Association Rules found by FP-Growth algorithm.

In Figure 5.8, we can observe the distribution of confidence values for the refined association rules generated during the

FP-Growth analysis. The histogram reveals a relatively broad distribution of confidence values, with many rules concentrated around the 0.4–0.6 range. This indicates that, for a substantial portion of the discovered rules, the antecedent moderately predicts the occurrence of the consequent. Such rules still provide useful information, especially in complex socioeconomic datasets where relationships are rarely deterministic.

In addition, the figure displays a noticeable concentration of rules with very high confidence values, approaching 0.8–0.9. These rules are particularly important because they indicate strong predictive reliability, meaning that the presence of the antecedent is highly associated with the occurrence of the consequent. In the context of this analysis, these high-confidence rules may reveal especially robust structural relationships between wealth-generation patterns and national well-being indicators.

Despite the filtering criteria applied during rule generation, the distribution still spans a wide range of confidence values. This variability is beneficial because it allows the selection of rules according to different levels of certainty and interpretability depending on the analytical objective. Consequently, the figure demonstrates that the extracted rules include both moderately reliable associations and highly predictive structural patterns.

The following association rules represent some of the most relevant structural relationships identified through the FP-Growth analysis. These rules connect patterns of wealth accumulation, industrial specialization, and billionaire wealth characteristics with country-level socioeconomic indicators such as life expectancy, inflation behavior, education, and fiscal conditions. Each rule is evaluated using support, confidence, and lift metrics, allowing the analysis to assess not only how frequently these patterns occur, but also the strength and reliability of the associations. Together, these rules provide insights into potential structural trade-offs and recurring socioeconomic dynamics embedded within different models of wealth generation.

2) Relevant Association Rules Found

TABLE 5.1
RELEVANT ASSOCIATION RULES FOUND WITH FP-GROWTH ALGORITHM

Rule	Antecedent	Consequent	Lift	Confidence
1	share_manufacturing_and_industrial_source_bin_is_low	cpi_change_bin_is_stable life_expectancy_bin_is_high	2.75	0.8
2	share_other_bin_is_high share_technology_source_bin_is_high	gross_tertiary_education_enrollment_bin_is_low	2.6	0.89
3	share_manufacturing_and_industrial_source_bin_is_low	cpi_change_bin_is_stable	2.37	0.9
4	share_industrials_&_manufacturing_bin_is_low share_manufacturing_and_industrial_source_bin_is_low	cpi_change_bin_is_stable	2.34	0.89
5	share_manufacturing_and_industrial_source_bin_is_low	life_expectancy_bin_is_high	2.34	0.8
6	share_other_bin_is_high	life_expectancy_bin_is_low gross_tertiary_education_enrollment_bin_is_low cpi_change_bin_is_stable	2.32	0.5
7	share_other_bin_is_low	life_expectancy_bin_is_high	2.06	0.6
8	share_other_bin_is_high	gross_tertiary_education_enrollment_bin_is_low	2.01	0.69
9	share_industrials_&_manufacturing_bin_is_low selfMade_ratio_bin_is_mid_selfmade_ratio	cpi_change_bin_is_stable	1.97	0.75
10	share_consumer_discretionary_bin_is_high	cpi_change_bin_is_stable life_expectancy_bin_is_high	1.96	0.57
11	share_finance_and_investments_source_bin_is_high	tax_revenue_bin_is_mid cpi_change_bin_is_moderate	1.96	0.35
12	share_industrials_&_manufacturing_bin_is_low selfMade_ratio_bin_is_mid_selfmade_ratio	life_expectancy_bin_is_high	1.95	0.67
13	share_industrials_&_manufacturing_bin_is_low	cpi_change_bin_is_stable	1.93	0.73
14	share_other_bin_is_low selfMade_ratio_bin_is_mid_selfmade_ratio	cpi_change_bin_is_stable	1.92	0.73
15	share_consumer_discretionary_bin_is_high	life_expectancy_bin_is_high	1.88	0.64
16	selfMade_ratio_bin_is_high_selfmade_ratio	tax_revenue_bin_is_mid cpi_change_bin_is_moderate	1.88	0.33
17	selfMade_ratio_bin_is_mid_selfmade_ratio	total_tax_rate_bin_is_low life_expectancy_bin_is_high	1.87	0.31

Rule	Antecedent	Consequent	Lift	Confidence
18	share_industrials_and_manufacturing_bin_is_low	cpi_change_bin_is_stable life_expectancy_bin_is_high	1.83	0.53
19	share_finance_and_investments_source_bin_is_mid	total_tax_rate_bin_is_low cpi_change_bin_is_stable	1.83	0.35
20	selfMade_ratio_bin_is_mid_selfmade_ratio	total_tax_rate_bin_is_low cpi_change_bin_is_stable	1.82	0.35

- Rule 1: Low industrial dependence associated with inflation stability and high life expectancy

Confidence: 0.80
Lift: 2.75

This rule indicates that 80% of the countries with low dependence on manufacturing and industrial wealth simultaneously exhibit stable inflation and high life expectancy. The very high lift value confirms that this relationship appears far more frequently than would be expected by chance. The pattern suggests that less industrially concentrated economies may benefit from more diversified and service-oriented structures, which are often associated with stronger healthcare systems, lower environmental pressures, and greater macroeconomic stability.

- Rule 2: High technological presence and "other" sectors associated with low tertiary education enrollment

Confidence: 0.89
Lift: 2.60

This was one of the strongest rules identified in the analysis. A confidence of 89% means that this relationship is highly reliable across the dataset. The rule suggests that countries with strong technological wealth combined with poorly categorized economic sectors frequently show low tertiary education enrollment. This may reveal economies where technological wealth is concentrated among a small elite or driven by isolated industries that do not necessarily translate into broad educational development or inclusive human capital growth.

- Rule 3: Low industrial dependence related to inflation stability

Confidence: 0.90
Lift: 2.37

This rule has one of the highest confidence values in the entire analysis, meaning that 90% of countries with low industrial dependence experience stable inflation. The association suggests that economies less tied to industrial production may be less exposed to supply chain disruptions, energy price volatility, and production cost fluctuations that commonly affect inflation dynamics.

- Rule 4: Lower manufacturing presence associated with economic stability

Confidence: 0.89
Lift: 2.34

This rule reinforces the previous pattern by combining two indicators of low industrial concentration. The confidence level of 89% shows that the rule is highly consistent. Together, these variables strongly suggest that countries with reduced industrial dependence tend to maintain more stable price dynamics, possibly due to more diversified or resilient economic structures.

- Rule 5: Low industrial orientation linked to higher life expectancy

Confidence: 0.80
Lift: 2.34

In 80% of the observed cases, countries with low industrial wealth concentration also presented high life expectancy. The strong lift value indicates that this is not a random coincidence. This relationship may reflect the long-term social advantages of economies more focused on services, technology, or knowledge-intensive sectors rather than environmentally intensive industrial activities.

- Rule 6: High presence of "other" sectors associated with low life expectancy and low tertiary education enrollment

Confidence: 0.50
Lift: 2.32

Unlike the previous rules with very high consistency, this association has a moderate confidence level, meaning that the pattern is present in only half of the countries that satisfy the antecedent. Nevertheless, the lift remains remarkably high, which makes the rule statistically important. This indicates that countries where wealth is concentrated in poorly defined or heterogeneous sectors are considerably more likely to experience both weaker educational access and lower life expectancy than expected under normal conditions. Rather than representing a deterministic relationship, the rule points toward a structural tendency linked to lower institutional development, economic informality, or limited specialization in high-value industries.

- Rule 7: Low presence of "other" sectors associated with economic and social stability

Confidence: 0.60
Lift: 2.06

This rule presents a more balanced behavior: the confidence is neither extremely high nor weak, suggesting that the relationship appears with reasonable regularity across the dataset. The result implies that countries with more clearly defined economic structures tend to achieve a combination of macroeconomic stability and favorable social indicators. The lift above 2 strengthens this interpretation, since the coexistence of stable inflation and high life expectancy occurs much more frequently in these countries than would normally be expected. In practical terms, economic specialization and institutional organization may contribute to more sustainable long-term development conditions.

- Rule 8: High "other" sector participation associated with low tertiary education enrollment

Confidence: 0.69
Lift: 2.01

Here the confidence becomes considerably stronger, approaching 70%, which means the rule is relatively reliable across the observed countries. The association suggests that economies dominated by poorly categorized or less specialized sectors frequently lack strong tertiary education systems. One possible explanation is that these economic structures may rely less on highly qualified labor, reducing both incentives and institutional investment in higher education. The lift confirms that the relationship is not random, but rather a recurring structural pattern within the dataset.

- Rule 9: Moderate self-made wealth and low industrialization associated with inflation stability

Confidence: 0.75
Lift: 1.97

This rule combines a high confidence level with a strong lift, making it both reliable and statistically relevant. Three out of four countries that satisfy the antecedent also present stable inflation. The result may characterize economies where entrepreneurial wealth generation exists in a balanced manner, without excessive concentration or dependence on traditional industrial production. Such environments could benefit from greater economic flexibility, lower exposure to industrial supply shocks, and more stable internal demand dynamics, all of which contribute to inflation control.

- Rule 10: Consumer discretionary sectors associated with stable inflation and high life expectancy

Confidence: 0.57
Lift: 1.96

This rule suggests that countries with a strong concentration of consumer discretionary industries frequently combine favorable macroeconomic and social conditions. Although the confidence is moderate, meaning the pattern is not universal, the lift close to 2 indicates that the association is still significantly stronger than random expectation. Economies driven by discretionary consumption are usually linked to higher purchasing power, stronger middle classes, and more mature service sectors, all of which may contribute to both economic stability and higher quality of life.

- Rule 11: High financial sector participation associated with moderate inflation and medium tax revenue

Confidence: 0.35
Lift: 1.96

This rule has relatively low confidence, indicating that the pattern is only present in about one-third of the countries satisfying the antecedent. However, the lift remains high, showing that the association still occurs far more often than expected by chance. One interpretation is that economies strongly oriented toward finance and investments tend to generate dynamic market activity and continuous capital movement, conditions that may contribute to moderate inflationary pressure while maintaining balanced fiscal revenues rather than extremely high or low taxation structures.

- Rule 12: Moderate self-made wealth and low manufacturing associated with high life expectancy

Confidence: 0.67
Lift: 1.95

The confidence of 67% shows that this relationship appears consistently across a majority of the observed countries. The rule suggests that economies with moderate entrepreneurial wealth generation and lower dependence on manufacturing tend to achieve better health and social outcomes. This combination may represent relatively balanced economic systems where innovation and entrepreneurship coexist with stable living conditions and reduced exposure to industrial environmental risks.

- Rule 13: Manufacturing concentration associated with inflation stability

Confidence: 0.73
Lift: 1.93

With a confidence above 70%, this rule can be considered highly reliable within the dataset. The result reinforces a recurring trend identified throughout the analysis: countries with lower manufacturing concentration frequently maintain more stable inflation. Reduced dependence on industrial production may lower exposure to supply chain disruptions, commodity shocks, and production-cost volatility, contributing to steadier price behavior.

- Rule 14: Low “other” sector participation and moderate self-made wealth associated with inflation stability

Confidence: 0.73
Lift: 1.92

This rule combines a strong confidence level with a relatively high lift, making it both reliable and statistically meaningful. Countries with more clearly structured economies and moderate levels of self-made wealth appear particularly associated with inflation stability. The result may indicate that balanced entrepreneurial ecosystems operating within organized economic structures are more capable of sustaining predictable macroeconomic conditions over time.

- Rule 15: Consumer discretionary sectors associated with high life expectancy

Confidence: 0.64
Lift: 1.88

This association indicates that countries with strong consumer discretionary sectors tend to present high life expectancy in nearly two-thirds of the observed cases. Consumer-oriented economies are commonly associated with higher income levels, better access to healthcare and services, and stronger domestic demand. The lift suggests that the relationship is not merely coincidental, but instead reflects broader socioeconomic conditions linked to prosperity and quality of life.

- Rule 16: High self-made wealth associated with moderate inflation and medium tax revenue

Confidence: 0.33
Lift: 1.88

Despite the low confidence, the lift indicates that this pattern still appears substantially more often than expected by chance. Countries with high proportions of self-made wealth may represent dynamic entrepreneurial economies with elevated investment activity and rapid market circulation. These conditions can stimulate economic growth but may also create moderate inflationary pressures while maintaining intermediate fiscal revenue levels instead of strongly redistributive tax systems.

- Rule 17: Moderate self-made wealth associated with low taxation and high life expectancy

Confidence: 0.31
Lift: 1.87

The confidence here is relatively weak, meaning the relationship is not broadly generalized across all countries in the dataset. Nevertheless, the lift close to 2 shows that the combination still occurs much more frequently than expected under random conditions. One possible interpretation is that countries with balanced entrepreneurial wealth generation may achieve favorable living conditions without relying on particularly high taxation levels, potentially reflecting efficient economic structures or productive service-based economies.

- Rule 18: Low industrial concentration associated with inflation stability and high life expectancy

Confidence: 0.53
Lift: 1.83

This rule presents moderate confidence but still reveals a meaningful structural tendency. Countries with lower manufacturing concentration frequently exhibit both stable inflation and strong social indicators simultaneously. The result further strengthens the broader pattern observed throughout the analysis, where diversified or post-industrial economies appear more strongly associated with stable and favorable socioeconomic environments.

- Rule 19: Medium financial sector participation associated with low taxes and stable inflation

Confidence: 0.35
Lift: 1.83

The relatively low confidence suggests that this association should not be interpreted as a dominant pattern across all countries. However, the lift still demonstrates statistical relevance. Economies with moderate (rather than extremely high) financial sector participation may benefit from a balance between investment activity and macroeconomic control, contributing to stable inflation while maintaining competitive tax environments.

- Rule 20: Moderate self-made wealth associated with low taxes and inflation stability

Confidence: 0.35
Lift: 1.82

Although the confidence is limited, the rule still appears more frequently than random expectation according to the lift value. The association may characterize economies where entrepreneurial activity develops in relatively balanced conditions, avoiding both excessive concentration of wealth and excessive fiscal pressure. Such environments could favor macroeconomic predictability and moderate taxation simultaneously.

VI. Clustering Task

A) K-Means clustering and Structural wealth ecosystem

Clustering techniques were applied to identify structural patterns among countries based on wealth generation characteristics. Unlike supervised learning, clustering does not require predefined labels, allowing the discovery of naturally emerging groups within the data. The objective of this analysis is to identify whether countries tend to organize into structurally distinct wealth ecosystems associated with innovation-driven economies, finance-oriented economies, or resource-dependent economies.

Country-Level Feature Engineering

Since the original dataset contains billionaire-level observations, a transformation process was required to construct country-level representations suitable for clustering analysis.First, billionaires were grouped by country, and aggregate structural indicators were computed for each nation. Countries with fewer than three billionaires were excluded in order to reduce statistical instability and avoid unreliable structural representations.

Next, sectoral proportions were calculated to estimate the relative importance of each macro-sector within each country. These features include the proportion of billionaires associated with technology, finance, manufacturing, energy, and other economic sectors.Additionally, the proportion of self-made billionaires was computed as an indicator of entrepreneurial wealth creation versus inherited wealth concentration.

To measure economic diversification, the Herfindahl-Hirschman Index (HHI) was calculated using sectoral proportions. Higher HHI values indicate stronger concentration of wealth in a limited number of industries, while lower values suggest more diversified economic structures.Finally, several socioeconomic indicators were incorporated, including tertiary education enrollment, life expectancy, tax revenue, GDP, and total population. These variables complement the wealth structure indicators by providing broader economic and demographic context.

TABLE 6.1
Summary of engineered features for contry

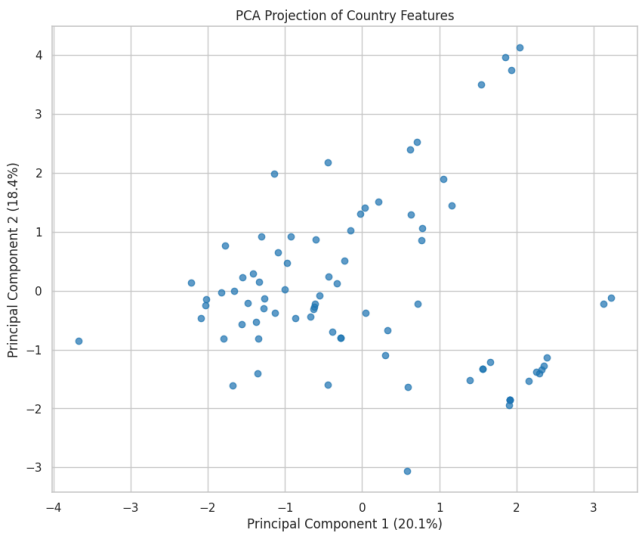
Feature	Description
tech_ratio	Proportion of billionaires in technology
finance_ratio	Proportion in finance
energy_ratio	Proportion in energy/mining
self_made_ratio	Ratio of self-made billionaires
hhi_index	Industrial concentration
billionaire_count	Number of billionaires
tertiary_education	Education enrollment
life_expectancy	Average life expectancy

Before applying clustering algorithms, all numerical features were standardized using z-score normalization. This step is necessary because clustering algorithms based on distance metrics, such as K-Means, are sensitive to differences in scale between variables. Standardization ensures that variables with larger numerical ranges do not dominate the clustering process.

We applied PCA to reduce the dimensionality of the feature space and facilitate visualization of structural relationships between countries. The original feature space, composed of approximately ten economic and social indicators, was projected into two principal components.

These components capture the largest proportion of variance present in the dataset, allowing visualization of country similarities in a reduced two-dimensional space.

Figure 6.0 PCA Projection of Countries Based on Structural Wealth Features

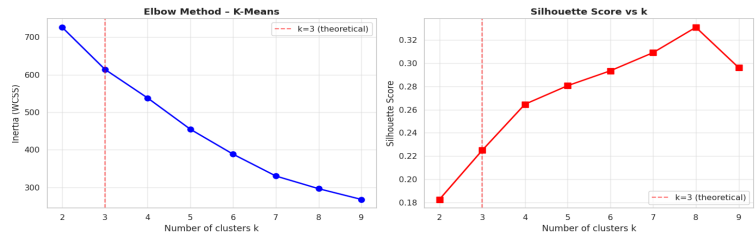


The Elbow Method was used to evaluate the optimal number of clusters by analyzing the reduction in within-cluster sum of squares (WCSS) as k increases.

Additionally, the Silhouette Score was computed to evaluate cluster cohesion and separation. Higher silhouette values indicate that observations are more similar to members of their own cluster than to observations from other clusters.

Both methods supported the use of k = 3, which also aligns with the theoretical framework proposed in this study.

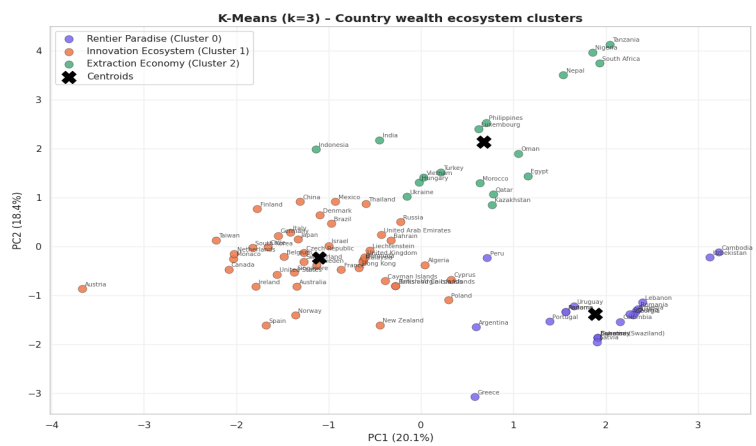
Figure 6.0 Elbow Method and Silhouette metrics



K-Means clustering was selected as the primary clustering method due to its interpretability and alignment with the theoretical hypothesis of three structural wealth ecosystems.

The algorithm partitions observations into k clusters by minimizing within-cluster variance, assigning each country to the nearest centroid according to Euclidean distance.

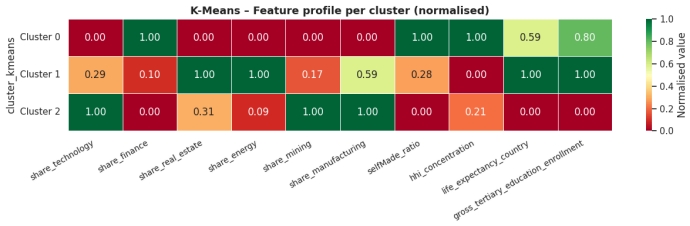
Figure 6.1 K-Means Clustering Results Projected onto the PCA Space



The normalized heatmap of cluster profiles allows comparison of structural characteristics across clusters. Higher intensity values indicate stronger representation of a given feature within a cluster.

The heatmap reveals clear structural differentiation between innovation-oriented economies, resource-dependent economies, and finance-dominated economies, supporting the hypothesis that distinct wealth ecosystems emerge across countries.

Figure 6.2 Heatmap of Normalized Cluster Profiles



B) Density-Based and Hierarchical Clustering Approaches

The second part of the clustering task focused on the application of two alternative unsupervised learning techniques: DBSCAN and Hierarchical Clustering using Ward linkage.

Unlike K-Means, these approaches allow a deeper structural interpretation of the dataset. DBSCAN identifies countries that behave as outliers and do not naturally belong to any cluster, while Hierarchical Clustering reveals the progressive similarity relationships between countries through a dendrogram representation.

The objective of these methods was to validate the theoretical three-archetype framework while also exploring whether atypical economic structures exist among countries

DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups observations according to their local density distribution.

One of the main advantages of DBSCAN is that it does not require defining the number of clusters in advance. Additionally, it can identify noise points or outlier countries that do not fit into the dominant economic structures.

For this analysis, the algorithm was applied using the following parameters:

- ε (epsilon) = 2.5
- min_samples = 3

The epsilon value was selected using a k-distance plot, where the elbow point indicated the optimal neighborhood radius.

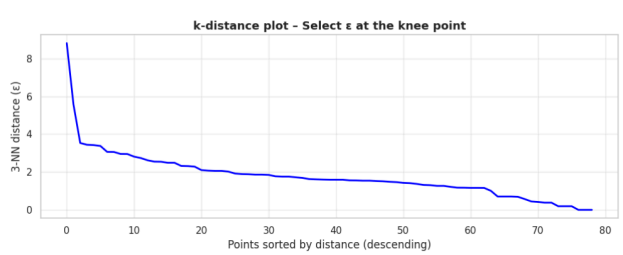


Figure 6.3 k-distance plot used to select ε for DBSCAN

After applying DBSCAN, several clusters and a set of outlier countries were identified. These outliers represent countries with highly specialized or structurally unique billionaire ecosystems.

The PCA projection shown in Figure X.X illustrates the spatial distribution of the detected clusters and noise points.

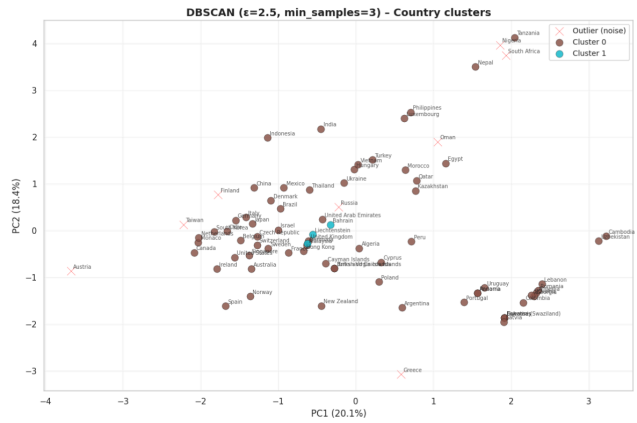


Figure 6.4 PCA visualization of DBSCAN clusters and outlier countries.

Countries identified as outliers included economies with strong dependence on sectors such as energy, mining, or finance. These countries exhibited high industrial concentration and lower diversification compared to the main clusters.

This result demonstrates the usefulness of DBSCAN for detecting atypical economic structures that traditional partition-based methods may fail to capture.

Hierarchical Clustering using Ward Linkage

Hierarchical Clustering was implemented using Ward linkage, a method that minimizes the variance within clusters during each merge step.

Unlike DBSCAN and K-Means, hierarchical clustering produces a dendrogram that visually represents the similarity relationships between countries at different levels of aggregation.

This approach provides a more interpretable understanding of how countries progressively group together according to their billionaire ecosystem structures.

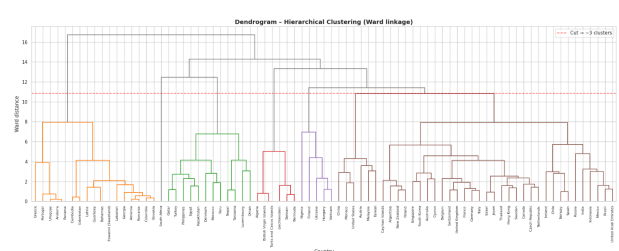


Figure 6.5 Dendrogram generated using Ward linkage.

The dendrogram suggested the existence of approximately three major clusters, partially supporting the original theoretical framework proposed in the project.

To validate the clustering quality, internal evaluation metrics were computed, including:

- Silhouette Score
- Davies-Bouldin Index
- Calinski-Harabasz Index

The results showed moderate agreement with the K-Means solution

Additionally, the Adjusted Rand Index (ARI) between Hierarchical Clustering and K-Means was calculated, obtaining a value close to 0.69, indicating moderate structural similarity between both methods.

Comparison of Clustering Algorithms

The three clustering approaches provided complementary insights into the global billionaire ecosystem structure.

K-Means aligned more directly with the predefined three-archetype theoretical framework. However, DBSCAN contributed by detecting countries with atypical economic compositions, while Hierarchical Clustering offered the most interpretable representation of country relationships through the dendrogram.

Although the Silhouette Scores remained relatively low across all methods, the clustering results still revealed meaningful structural patterns related to innovation, financial concentration, and resource dependence.

=== COMPARISON OF ALL THREE CLUSTERING ALGORITHMS ===

Algorithm	k / clusters found	Outliers detected	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑	Silhouette ≥ 0.50	DBI < 1.00
K-Means	3	0	0.2249	1.6968	15.75	✗	✗
DBSCAN	2	8	0.2401	0.8253	6.16	✗	✓
Hierarchical (Ward)	3	0	0.2278	1.5881	14.65	✗	✗

Figure 6.6 Comparison of clustering evaluation metrics

VII. Conclusions

The association rule mining task successfully uncovered meaningful patterns linking individual billionaire characteristics with country-level socioeconomic indicators. One of the most significant findings is the strong association between high tertiary education enrollment, a higher proportion of self-made billionaires, and sectors such as Technology and Finance & Investments.

Additionally, consistent rules were discovered connecting countries with high GDP and elevated life expectancy to a greater presence of billionaires in high-value industries and younger profiles. In contrast, traditional sources of wealth such as Real Estate and extractive industries showed stronger associations with environments characterized by lower entrepreneurial dynamism and a higher proportion of inherited wealth.

These findings confirm that the generation of extreme wealth is not random but is closely tied to the economic, educational, and demographic ecosystem of each country, providing empirical evidence of the structural factors that promote fortune creation.

The clustering analysis at the country level effectively identified three distinct wealth generation ecosystems:

- Innovation Ecosystem: Characterized by a high proportion of self-made billionaires, strong presence in Technology and Manufacturing sectors, younger average age, and high human capital indicators (particularly tertiary education). This cluster mainly includes the United States, China, and other dynamic economies.
- Rentier Paradise: Dominated by a higher share of inherited wealth, concentration in Finance, Real Estate, and diversified investments. These countries exhibit high macroeconomic stability but lower entrepreneurial dynamism.
- Extraction Economy: Marked by high sectoral concentration in energy, natural resources, and traditional manufacturing, along with lower diversification and generally weaker performance in education and innovation indicators.

This segmentation validates the project's conceptual hypothesis and demonstrates that countries can be meaningfully classified according to their structural model of extreme wealth creation. The results offer a valuable analytical framework for understanding global inequality and for designing policies aimed at fostering more innovative and dynamic economic ecosystems.

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